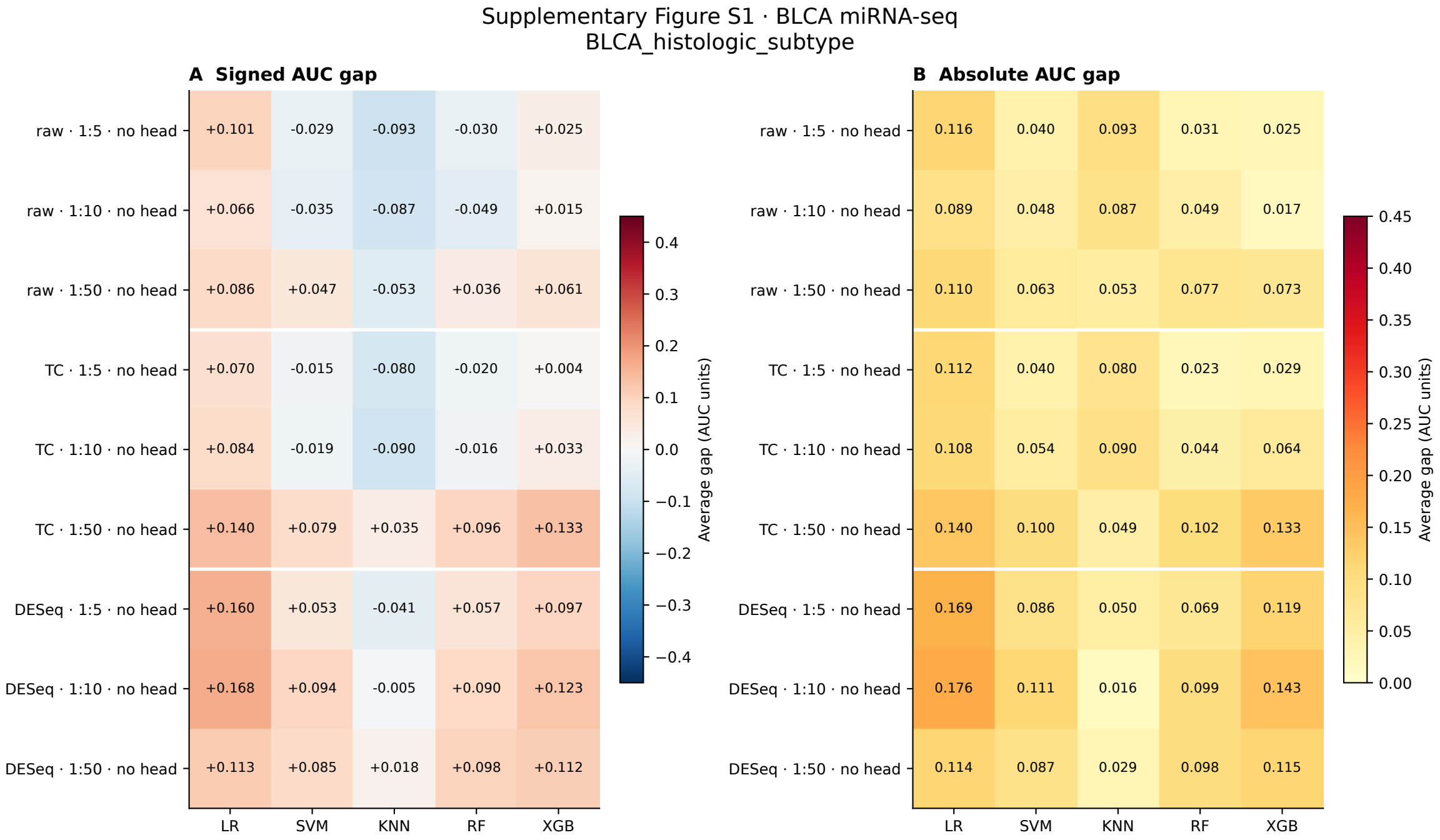
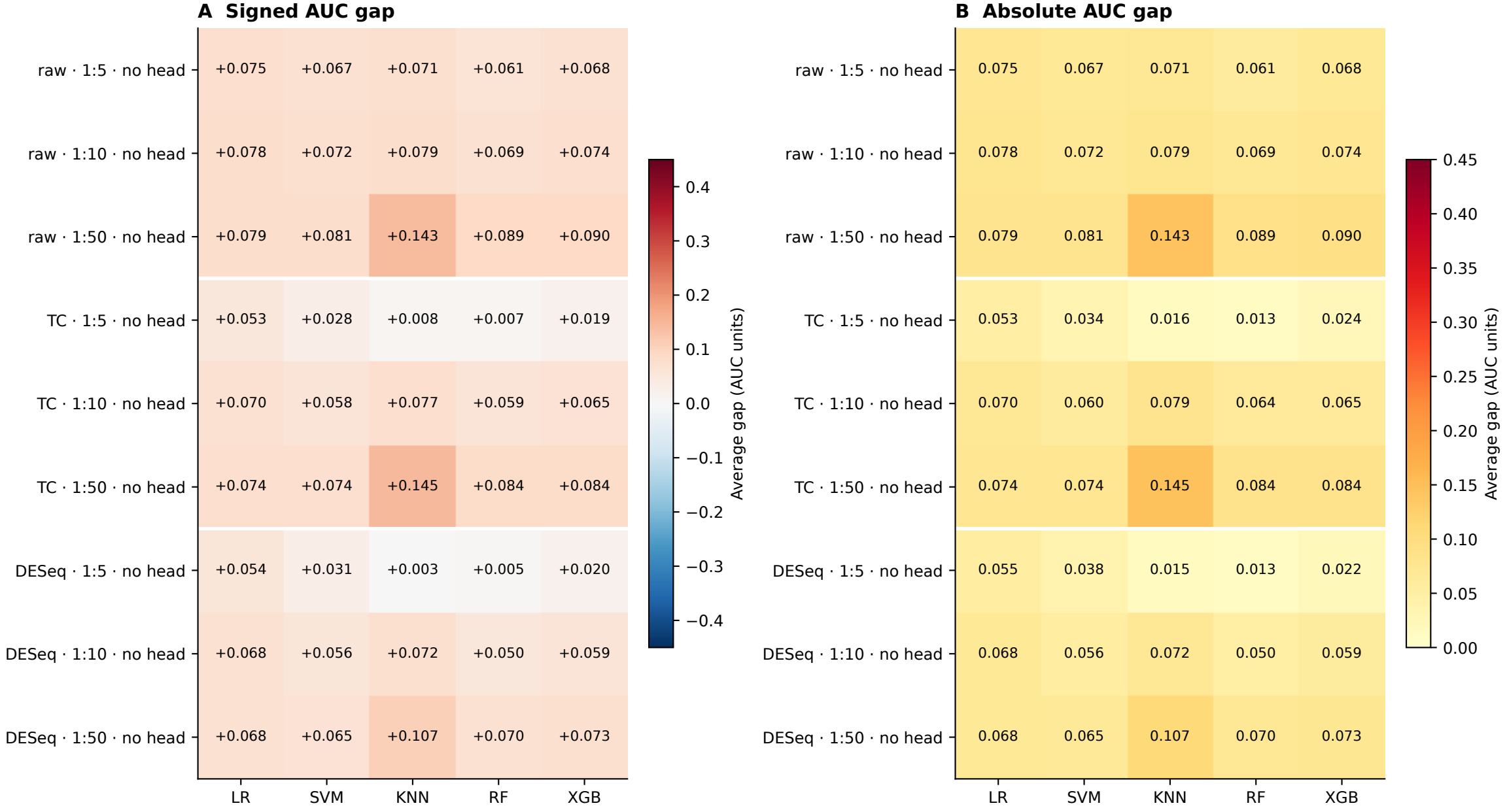
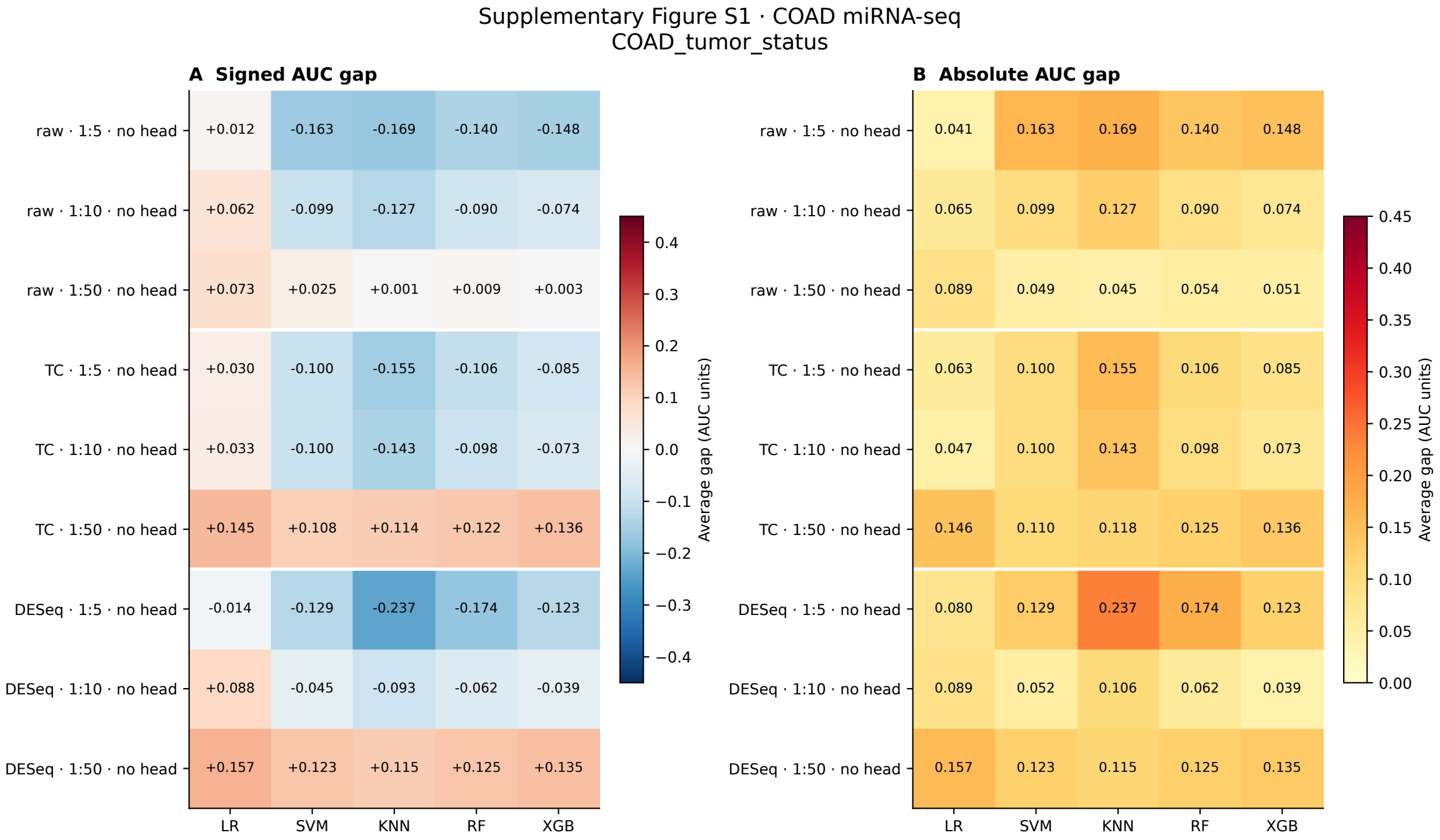
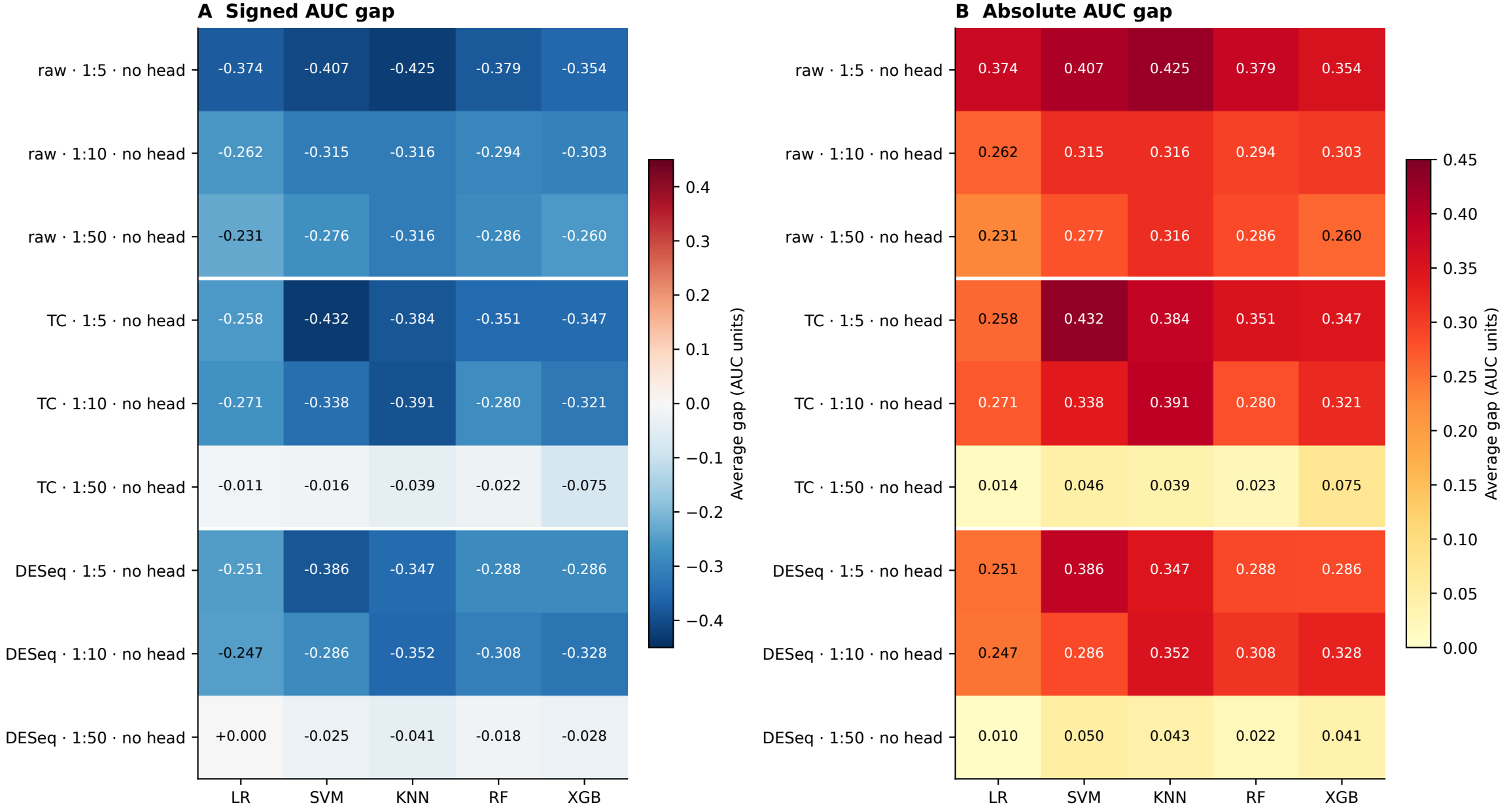

Supplementary Figure S1 · BLCA miRNA-seq
BLCA_histologic_subtype
A Signed AUC gap
B Absolute AUC gap
raw · 1:5 · no head
raw · 1:10 · no head
raw · 1:50 · no head
TC · 1:5 · no head
TC · 1:10 · no head
TC · 1:50 · no head
DESeq · 1:5 · no head
DESeq · 1:10 · no head
DESeq · 1:50 · no head
+0.101 -0.029 -0.093 -0.030 +0.025
+0.066 -0.035 -0.087 -0.049 +0.015
+0.086 +0.047 -0.053 +0.036 +0.061
+0.070 -0.015 -0.080 -0.020 +0.004
+0.084 -0.019 -0.090 -0.016 +0.033
+0.140 +0.079 +0.035 +0.096 +0.133
+0.160 +0.053 -0.041 +0.057 +0.097
+0.168 +0.094 -0.005 +0.090 +0.123
+0.113 +0.085 +0.018 +0.098 +0.112
0.116 0.040 0.093 0.031 0.025
0.089 0.048 0.087 0.049 0.017
0.110 0.063 0.053 0.077 0.073
0.112 0.040 0.080 0.023 0.029
0.108 0.054 0.090 0.044 0.064
0.140 0.100 0.049 0.102 0.133
0.169 0.086 0.050 0.069 0.119
0.176 0.111 0.016 0.099 0.143
0.114 0.087 0.029 0.098 0.115
LR SVM KNN RF XGB
Average gap (AUC units)
0.4 0.3 0.2 0.1 0.0 −0.1 −0.2 −0.3 −0.4
0.45 0.40 0.35 0.30 0.25 0.20 0.15 0.10 0.05 0.00

Supplementary Figure S1 · BRCA miRNA-seq
BRCA_breast_carcinoma_estrogen_receptor_status

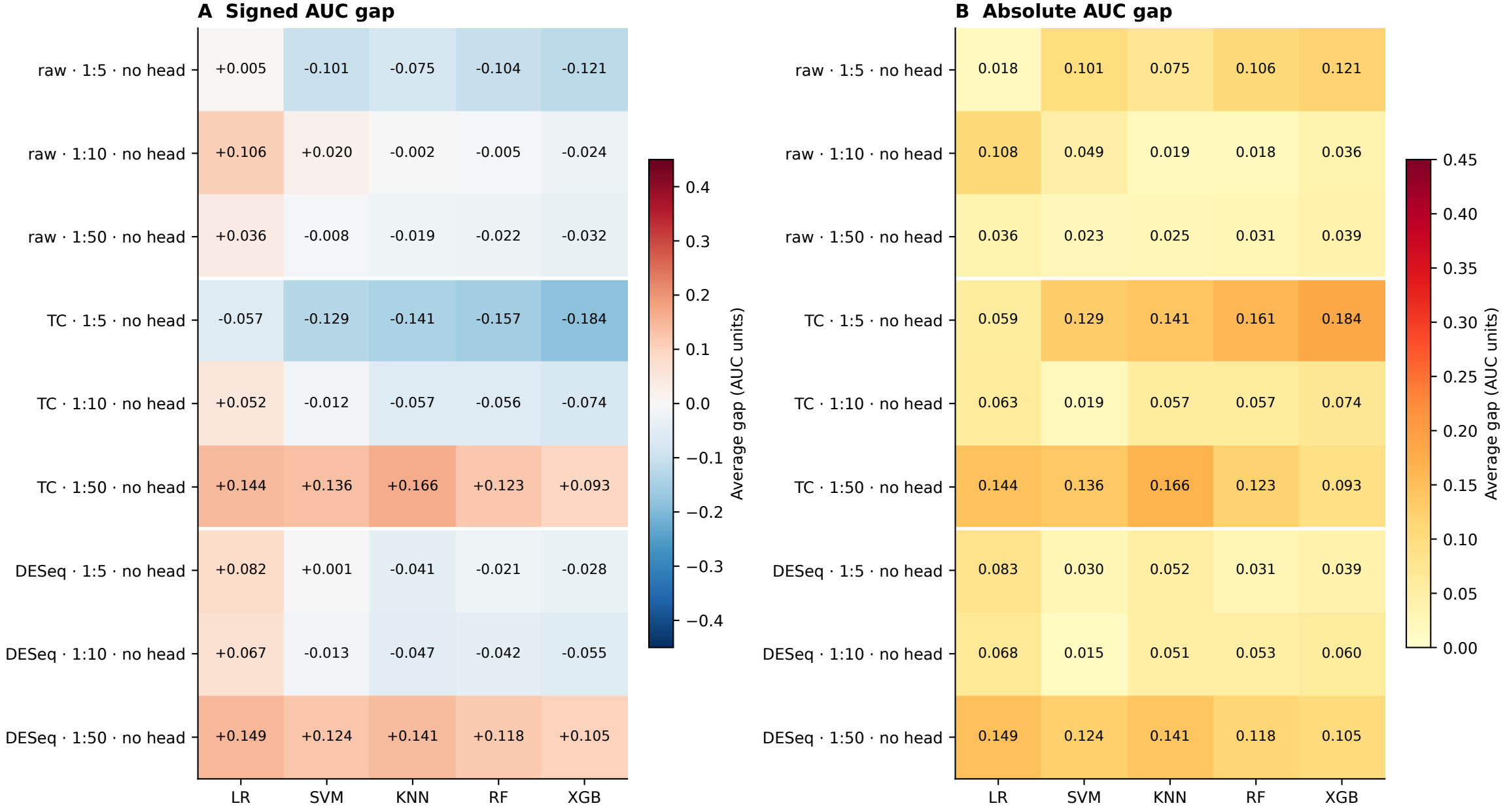


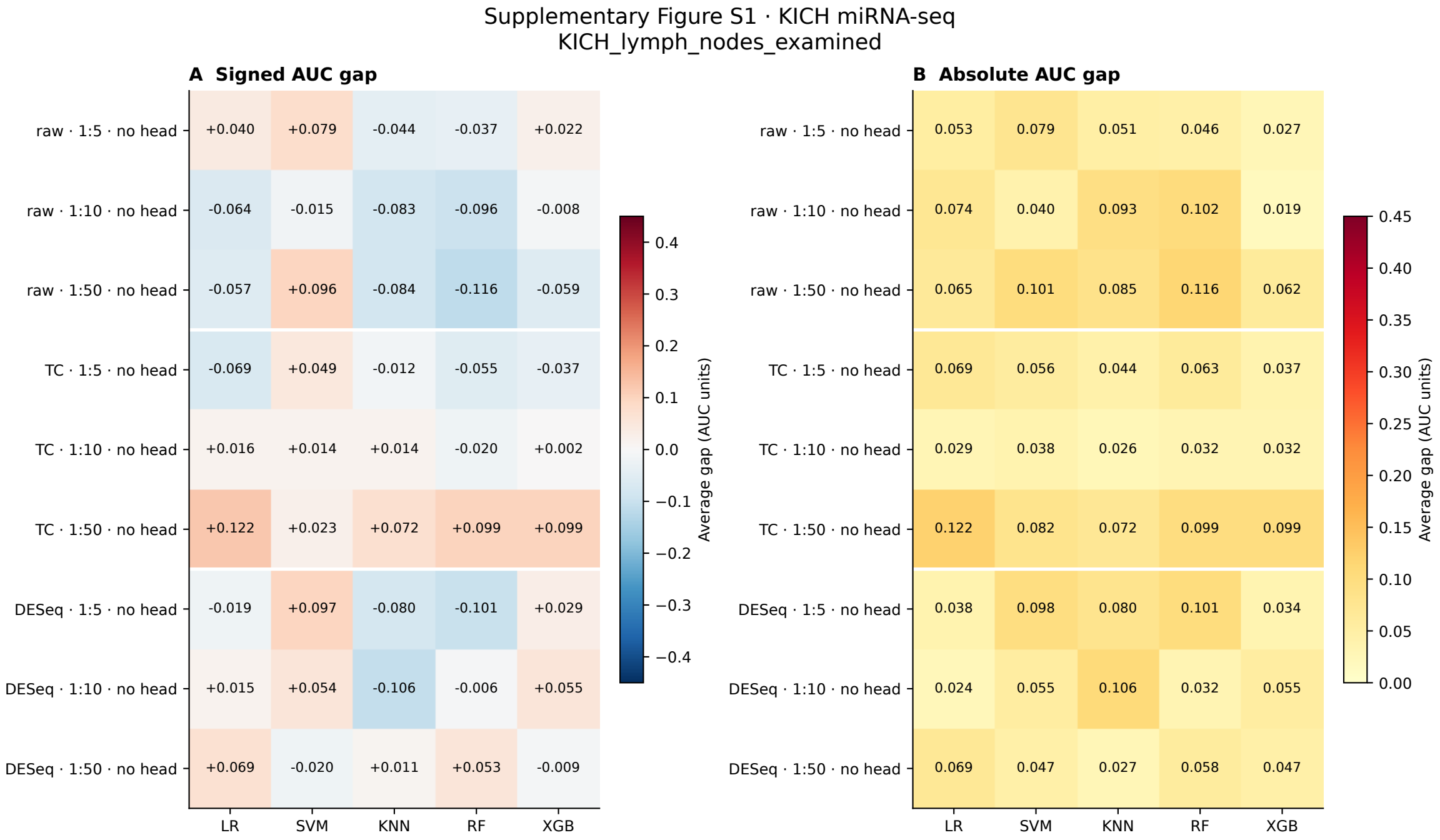

Supplementary Figure S1 · COAD miRNA-seq
COAD_tumor_status
A Signed AUC gap
B Absolute AUC gap
raw · 1:5 · no head
raw · 1:10 · no head
raw · 1:50 · no head
TC · 1:5 · no head
TC · 1:10 · no head
TC · 1:50 · no head
DESeq · 1:5 · no head
DESeq · 1:10 · no head
DESeq · 1:50 · no head
LR
SVM
KNN
RF
XGB
+0.012 -0.163 -0.169 -0.140 -0.148
+0.062 -0.099 -0.127 -0.090 -0.074
+0.073 +0.025 +0.001 +0.009 +0.003
+0.030 -0.100 -0.155 -0.106 -0.085
+0.033 -0.100 -0.143 -0.098 -0.073
+0.145 +0.108 +0.114 +0.122 +0.136
-0.014 -0.129 -0.237 -0.174 -0.123
+0.088 -0.045 -0.093 -0.062 -0.039
+0.157 +0.123 +0.115 +0.125 +0.135
0.041 0.163 0.169 0.140 0.148
0.065 0.099 0.127 0.090 0.074
0.089 0.049 0.045 0.054 0.051
0.063 0.100 0.155 0.106 0.085
0.047 0.100 0.143 0.098 0.073
0.146 0.110 0.118 0.125 0.136
0.080 0.129 0.237 0.174 0.123
0.089 0.052 0.106 0.062 0.039
0.157 0.123 0.115 0.125 0.135
Average gap (AUC units)
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
−0.4
Average gap (AUC units)
0.45
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00

Supplementary Figure S1 · ESCA miRNA-seq
ESCA_icd_o_3_histology

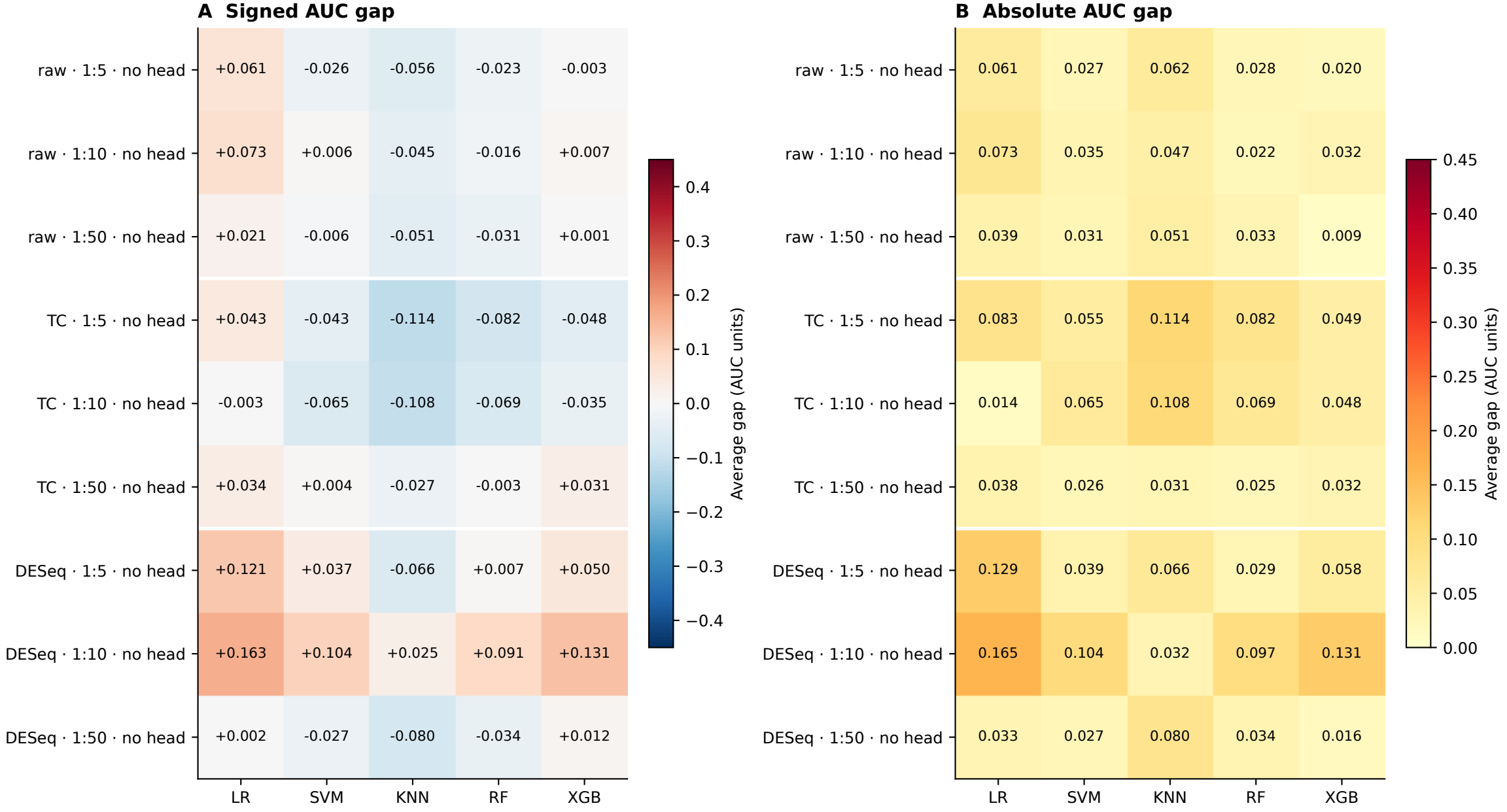


Supplementary Figure S1 · HNSC miRNA-seq
HNSC_anatomic_organ_subdivision

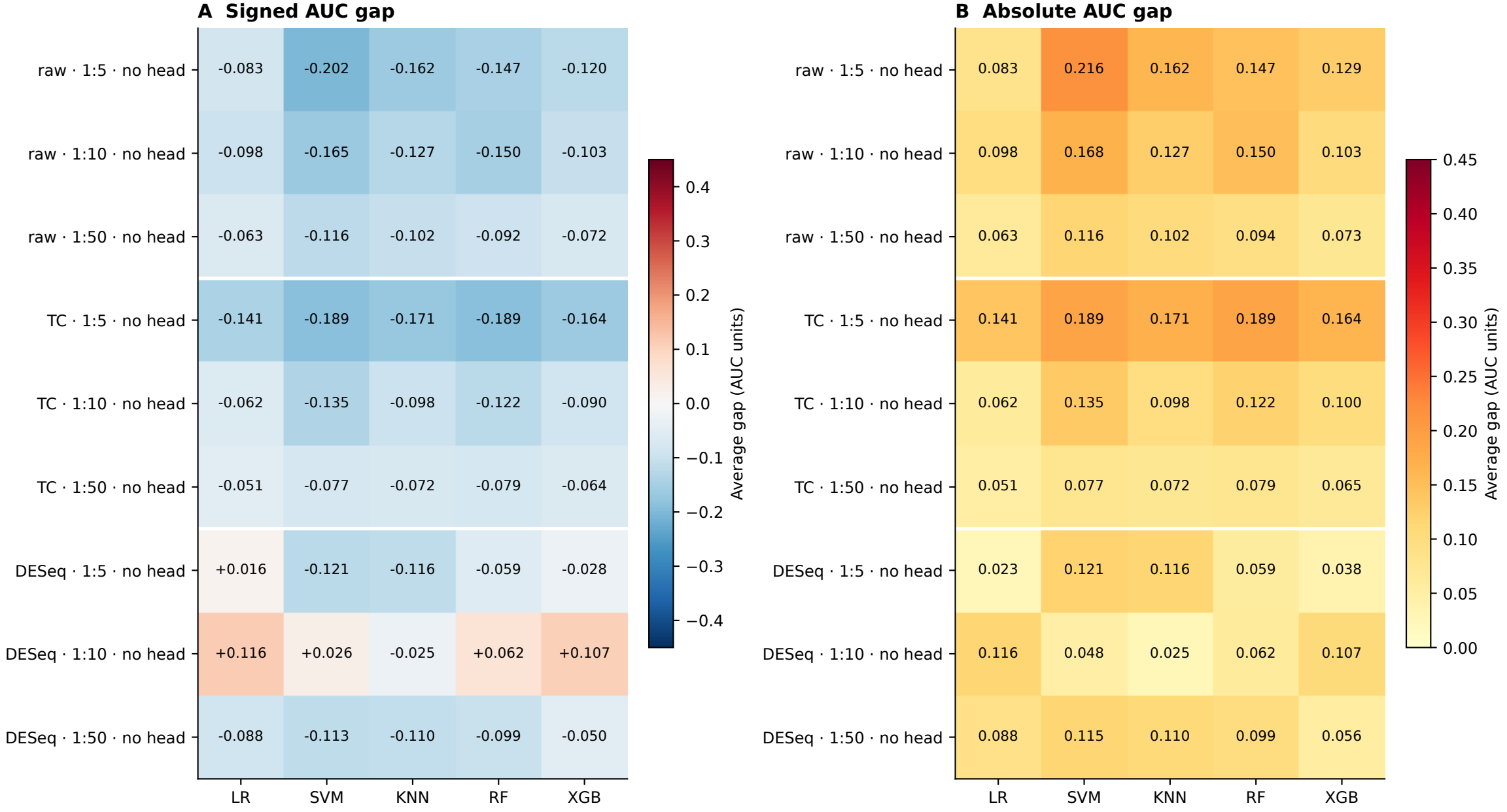



Supplementary Figure S1 · KICH miRNA-seq
KICH_lymph_nodes_examined
A Signed AUC gap
B Absolute AUC gap
raw · 1:5 · no head
raw · 1:10 · no head
raw · 1:50 · no head
TC · 1:5 · no head
TC · 1:10 · no head
TC · 1:50 · no head
DESeq · 1:5 · no head
DESeq · 1:10 · no head
DESeq · 1:50 · no head
+0.040 +0.079 -0.044 -0.037 +0.022
-0.064 -0.015 -0.083 -0.096 -0.008
-0.057 +0.096 -0.084 -0.116 -0.059
-0.069 +0.049 -0.012 -0.055 -0.037
+0.016 +0.014 +0.014 -0.020 +0.002
+0.122 +0.023 +0.072 +0.099 +0.099
-0.019 +0.097 -0.080 -0.101 +0.029
+0.015 +0.054 -0.106 -0.006 +0.055
+0.069 -0.020 +0.011 +0.053 -0.009
0.053 0.079 0.051 0.046 0.027
0.074 0.040 0.093 0.102 0.019
0.065 0.101 0.085 0.116 0.062
0.069 0.056 0.044 0.063 0.037
0.029 0.038 0.026 0.032 0.032
0.122 0.082 0.072 0.099 0.099
0.038 0.098 0.080 0.101 0.034
0.024 0.055 0.106 0.032 0.055
0.069 0.047 0.027 0.058 0.047
LR SVM KNN RF XGB
Average gap (AUC units)
0.4 0.3 0.2 0.1 0.0 −0.1 −0.2 −0.3 −0.4
0.45 0.40 0.35 0.30 0.25 0.20 0.15 0.10 0.05 0.00

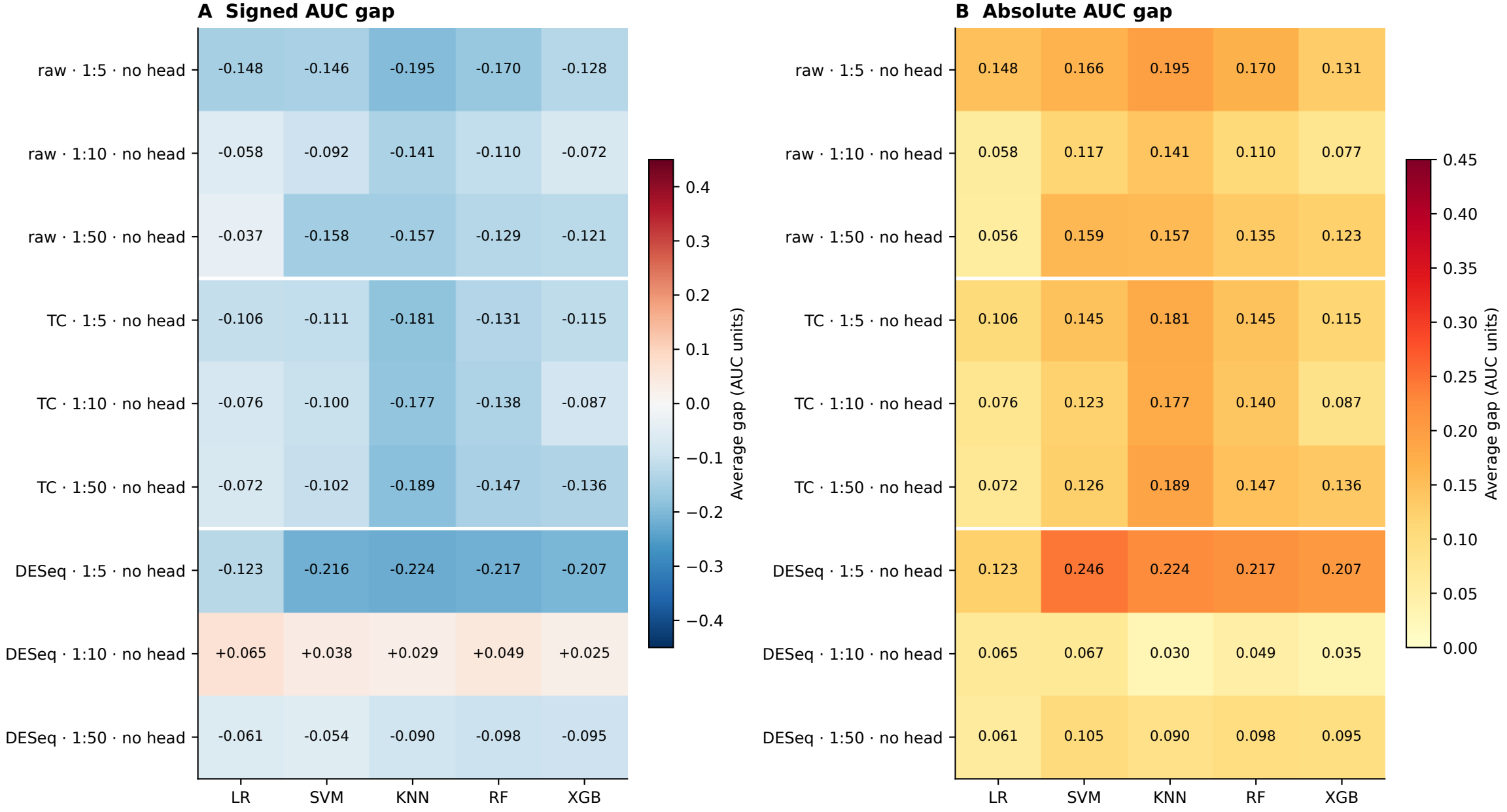
Supplementary Figure S1 · KIRC miRNA-seq
KIRC_ajcc_pathologic_tumor_stage

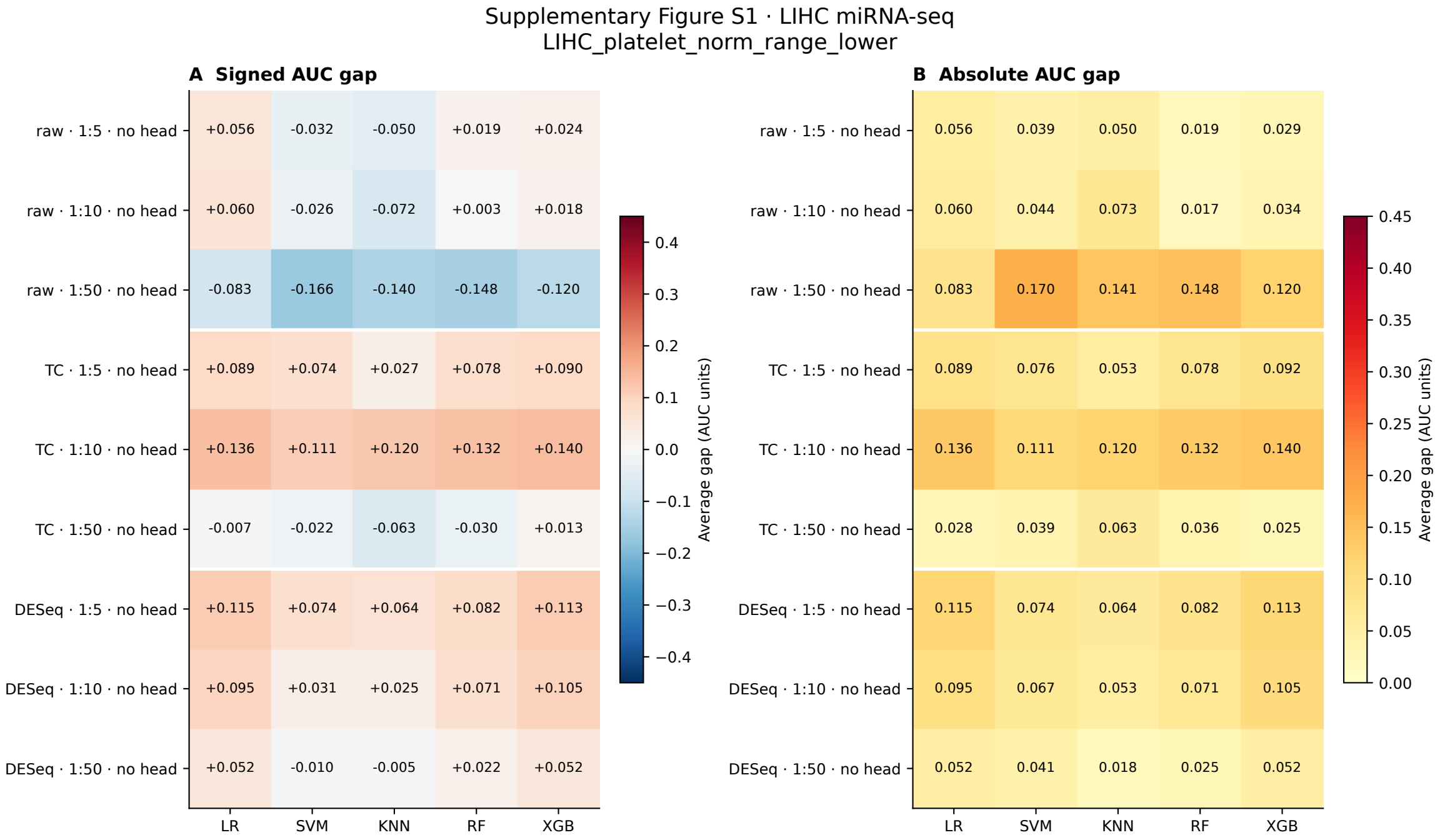


Supplementary Figure S1 · KIRP miRNA-seq
KIRP_ajcc_pathologic_tumor_stage

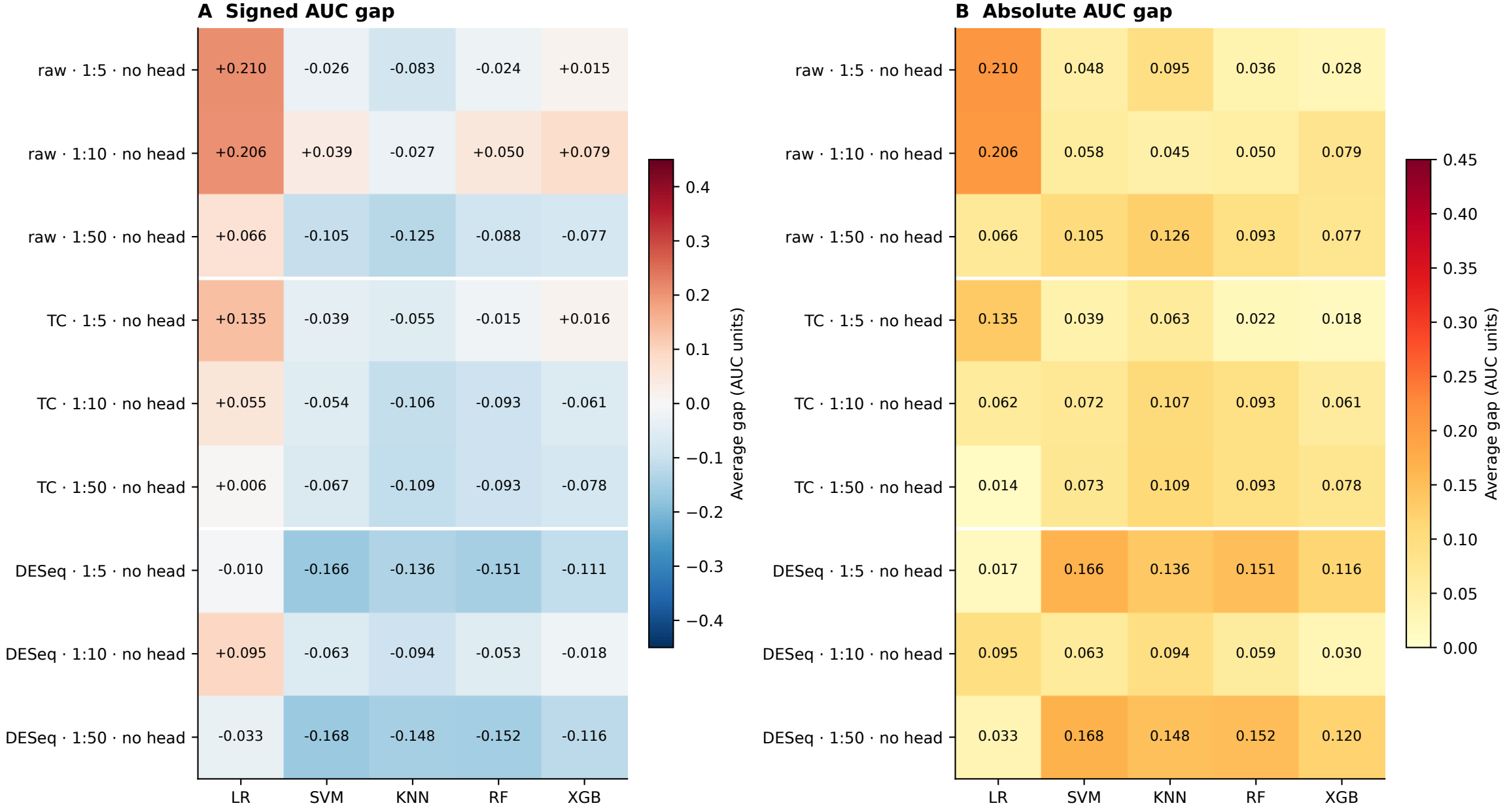


Supplementary Figure S1 · LAML miRNA-seq
LAML_wbc_24hr_of_banking

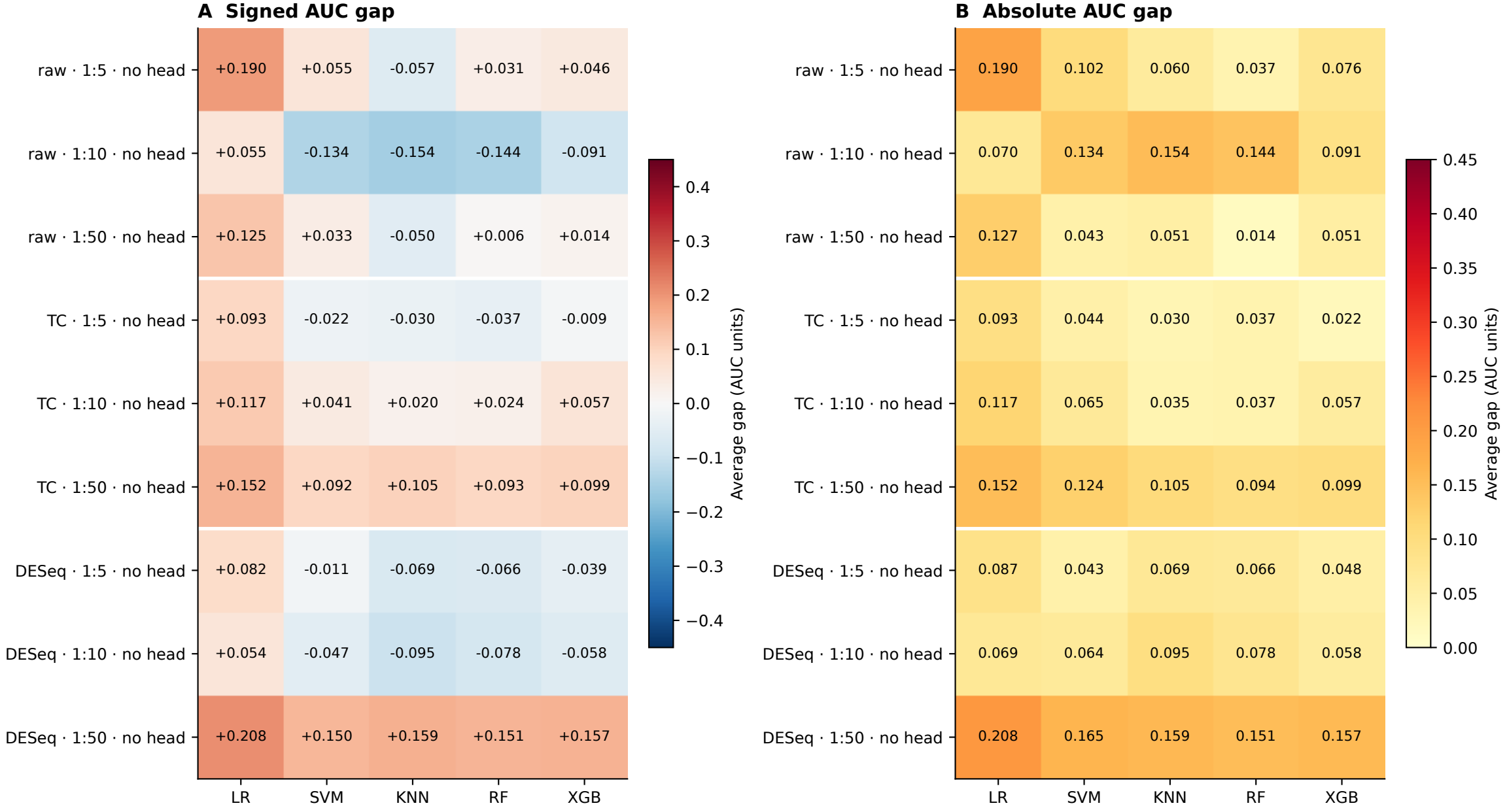



Supplementary Figure S1 · LIHC miRNA-seq
LIHC_platelet_norm_range_lower
A Signed AUC gap
B Absolute AUC gap
raw · 1:5 · no head
raw · 1:10 · no head
raw · 1:50 · no head
TC · 1:5 · no head
TC · 1:10 · no head
TC · 1:50 · no head
DESeq · 1:5 · no head
DESeq · 1:10 · no head
DESeq · 1:50 · no head
LR
SVM
KNN
RF
XGB
+0.056 -0.032 -0.050 +0.019 +0.024
+0.060 -0.026 -0.072 +0.003 +0.018
-0.083 -0.166 -0.140 -0.148 -0.120
+0.089 +0.074 +0.027 +0.078 +0.090
+0.136 +0.111 +0.120 +0.132 +0.140
-0.007 -0.022 -0.063 -0.030 +0.013
+0.115 +0.074 +0.064 +0.082 +0.113
+0.095 +0.031 +0.025 +0.071 +0.105
+0.052 -0.010 -0.005 +0.022 +0.052
0.056 0.039 0.050 0.019 0.029
0.060 0.044 0.073 0.017 0.034
0.083 0.170 0.141 0.148 0.120
0.089 0.076 0.053 0.078 0.092
0.136 0.111 0.120 0.132 0.140
0.028 0.039 0.063 0.036 0.025
0.115 0.074 0.064 0.082 0.113
0.095 0.067 0.053 0.071 0.105
0.052 0.041 0.018 0.025 0.052
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
−0.4
Average gap (AUC units)
0.45
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
Average gap (AUC units)

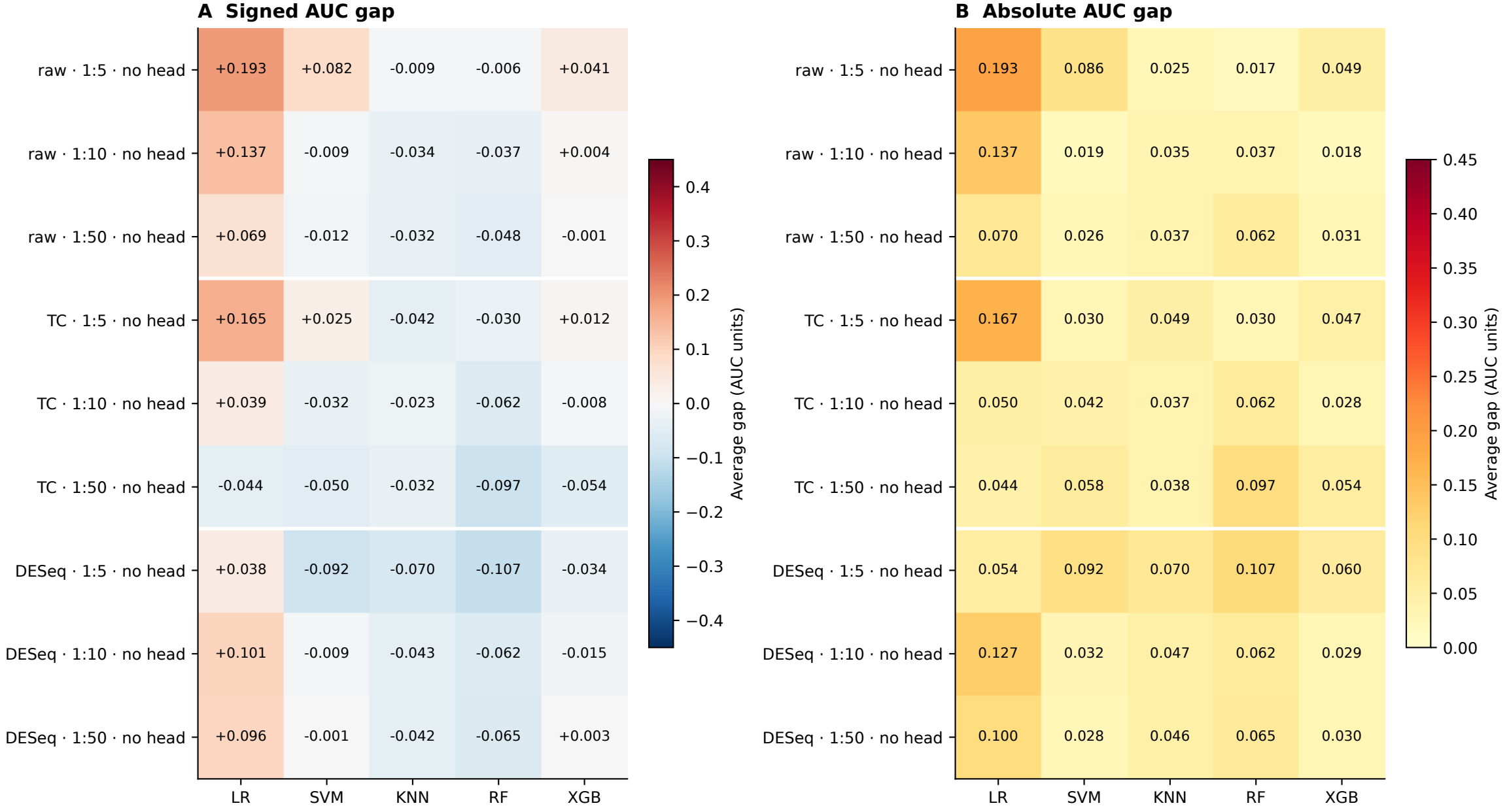
Supplementary Figure S1 · LUAD miRNA-seq
LUAD_initial_pathologic_dx_year



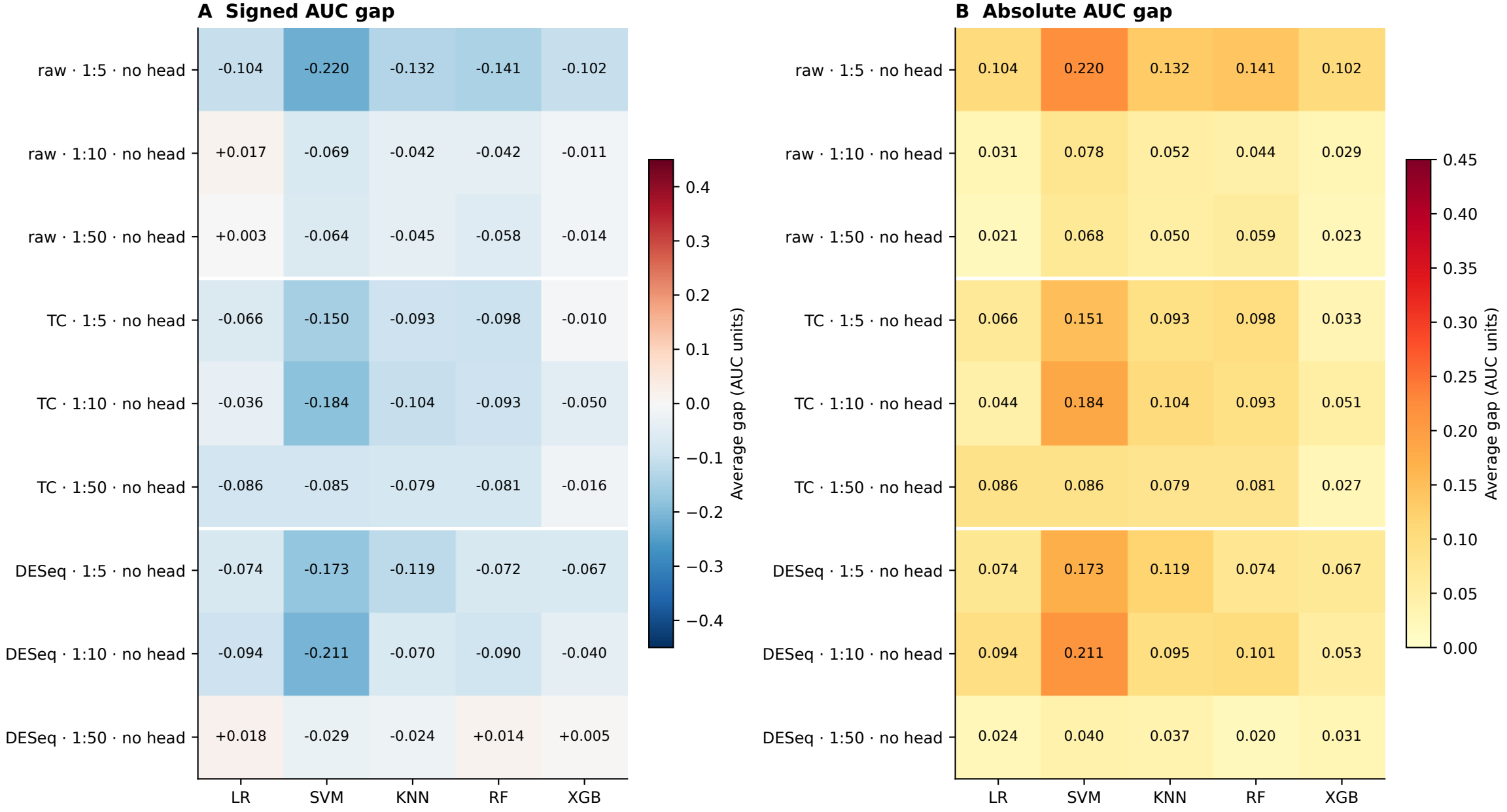
Supplementary Figure S1 · LUSC miRNA-seq
LUSC_initial_pathologic_dx_year



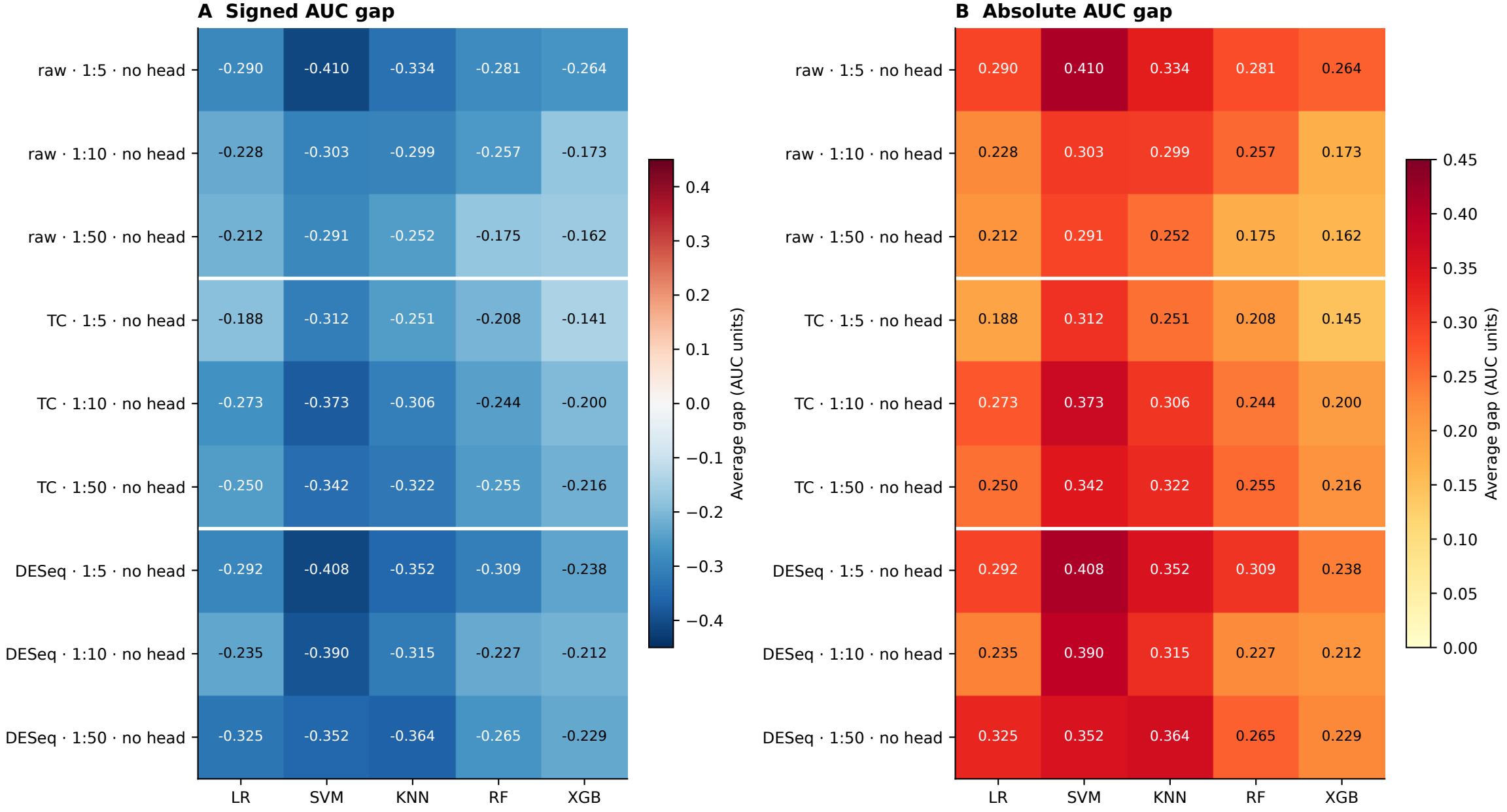
Supplementary Figure S1 · OV miRNA-seq
OV_tumor_residual_disease

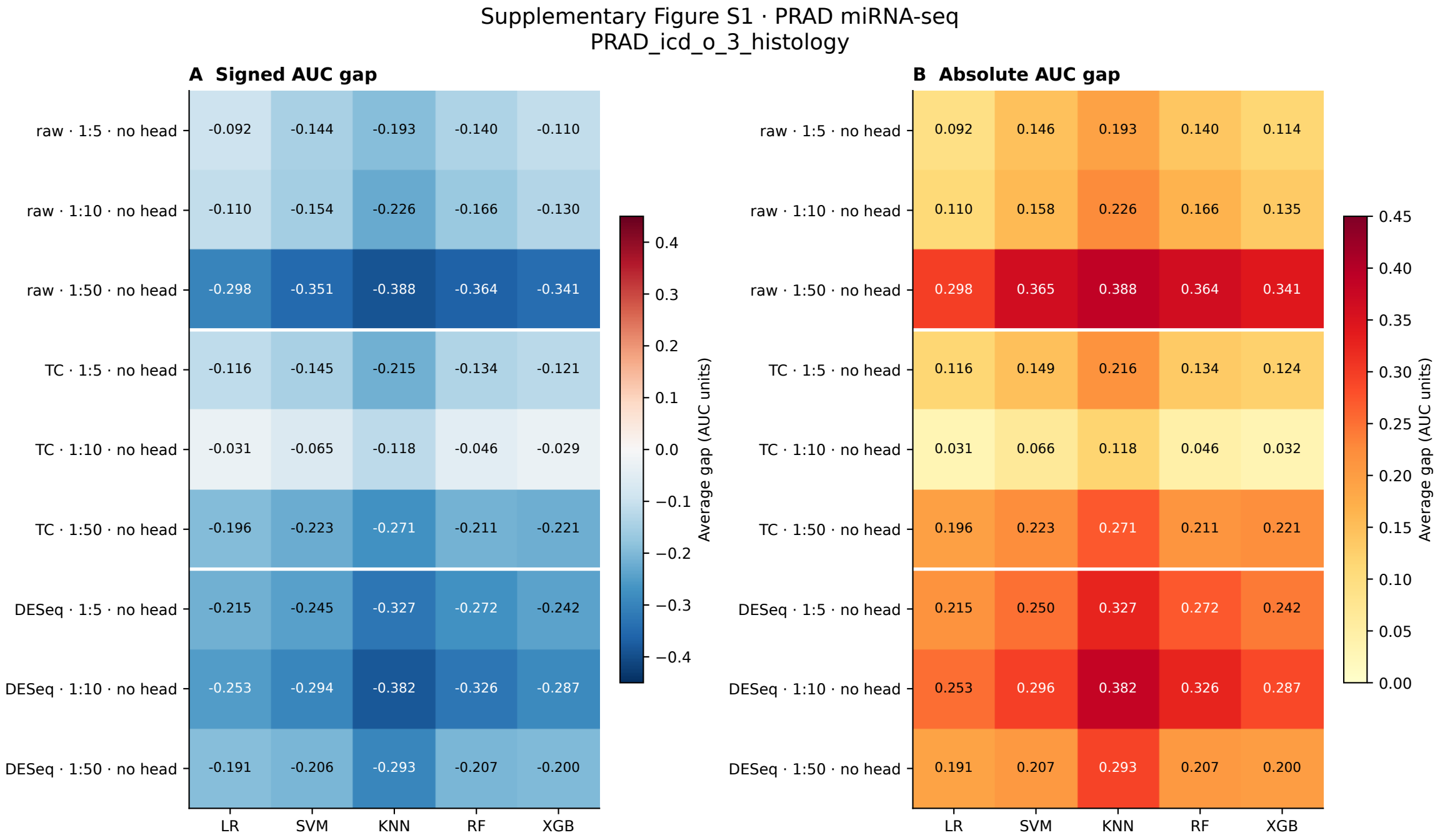


Supplementary Figure S1 · PAAD miRNA-seq
PAAD_anatomic_neoplasm_subdivision

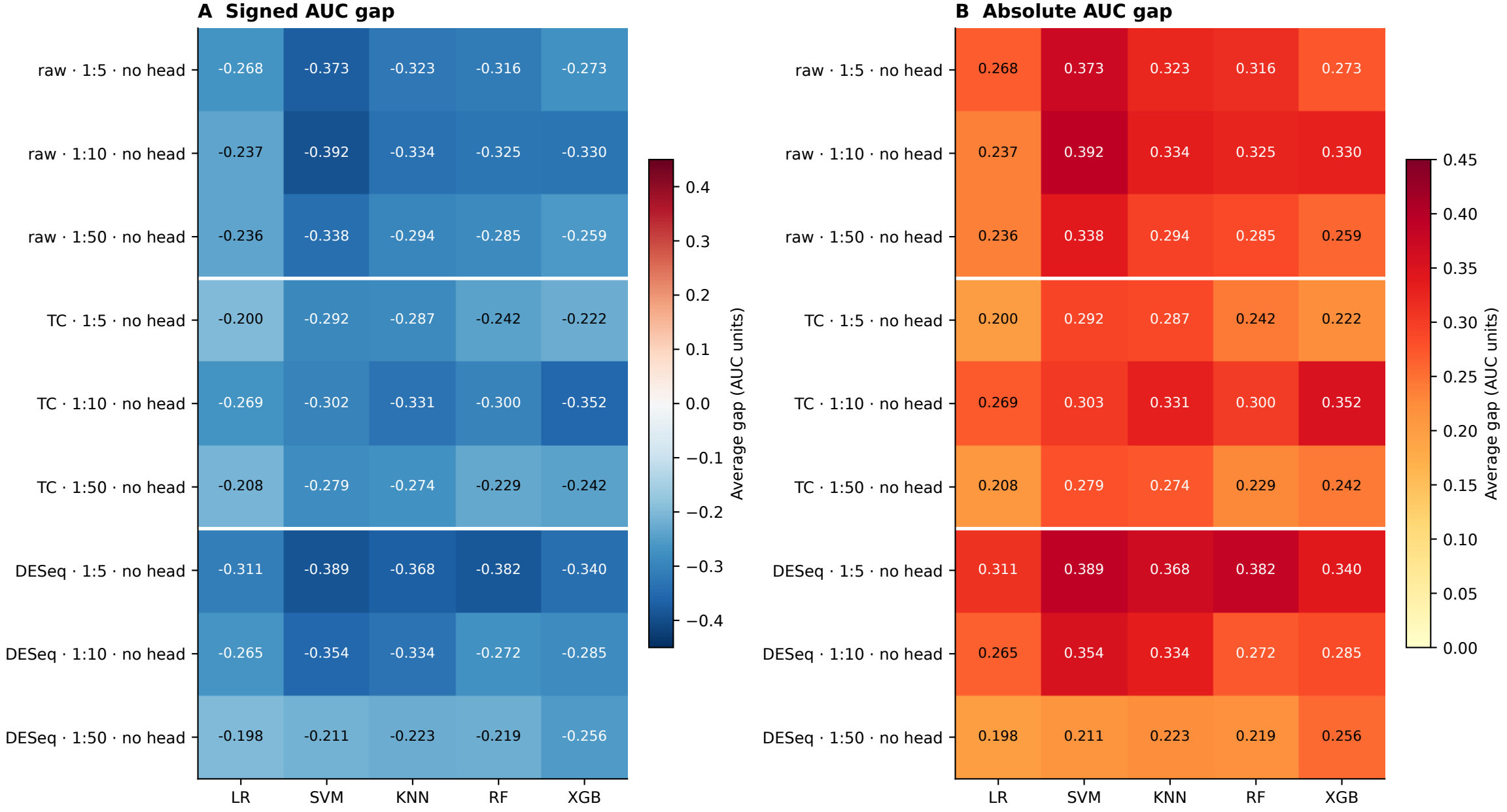


Supplementary Figure S1 · PCPG miRNA-seq
PCPG_primary_pathology_histological_type

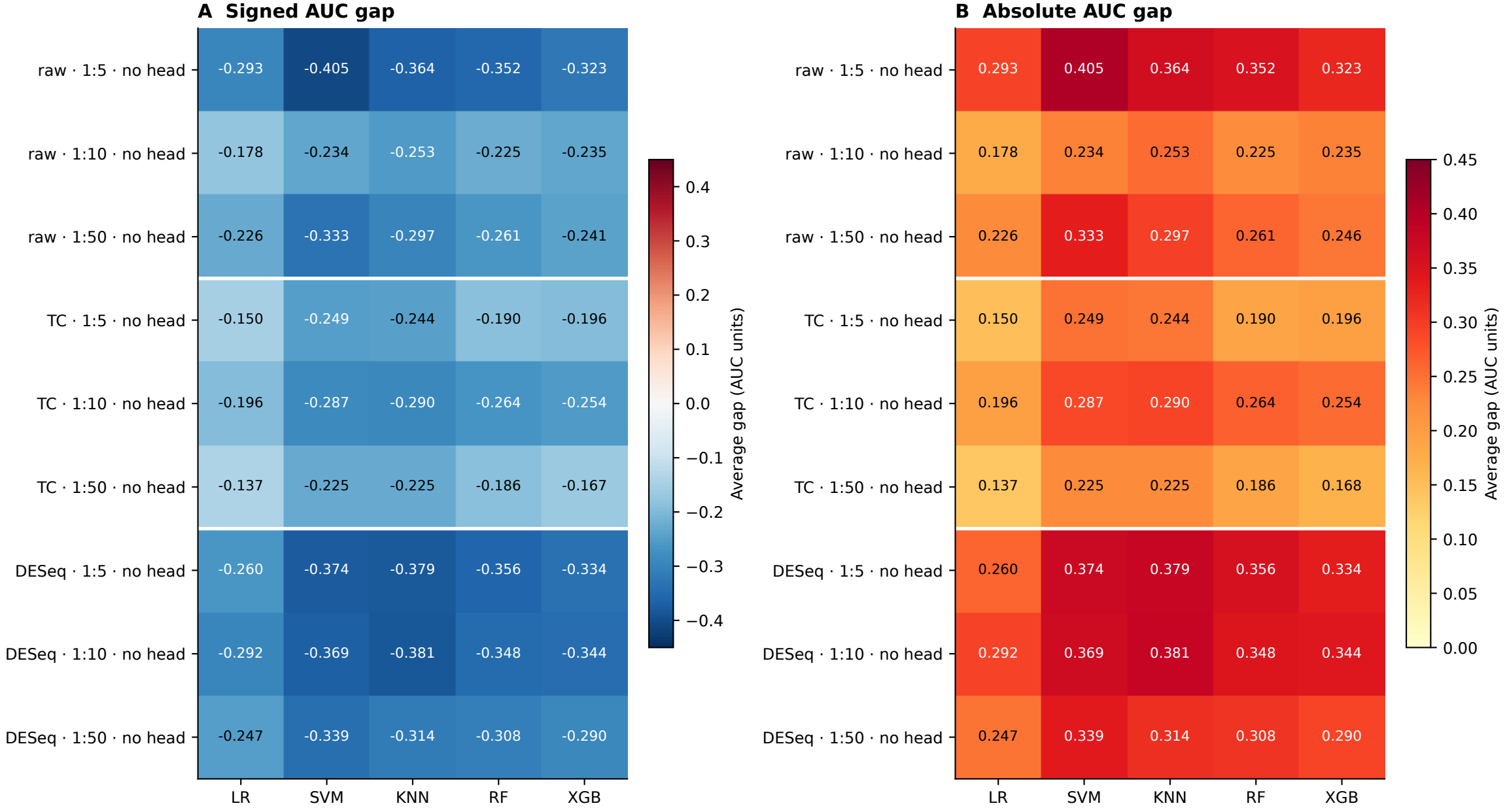


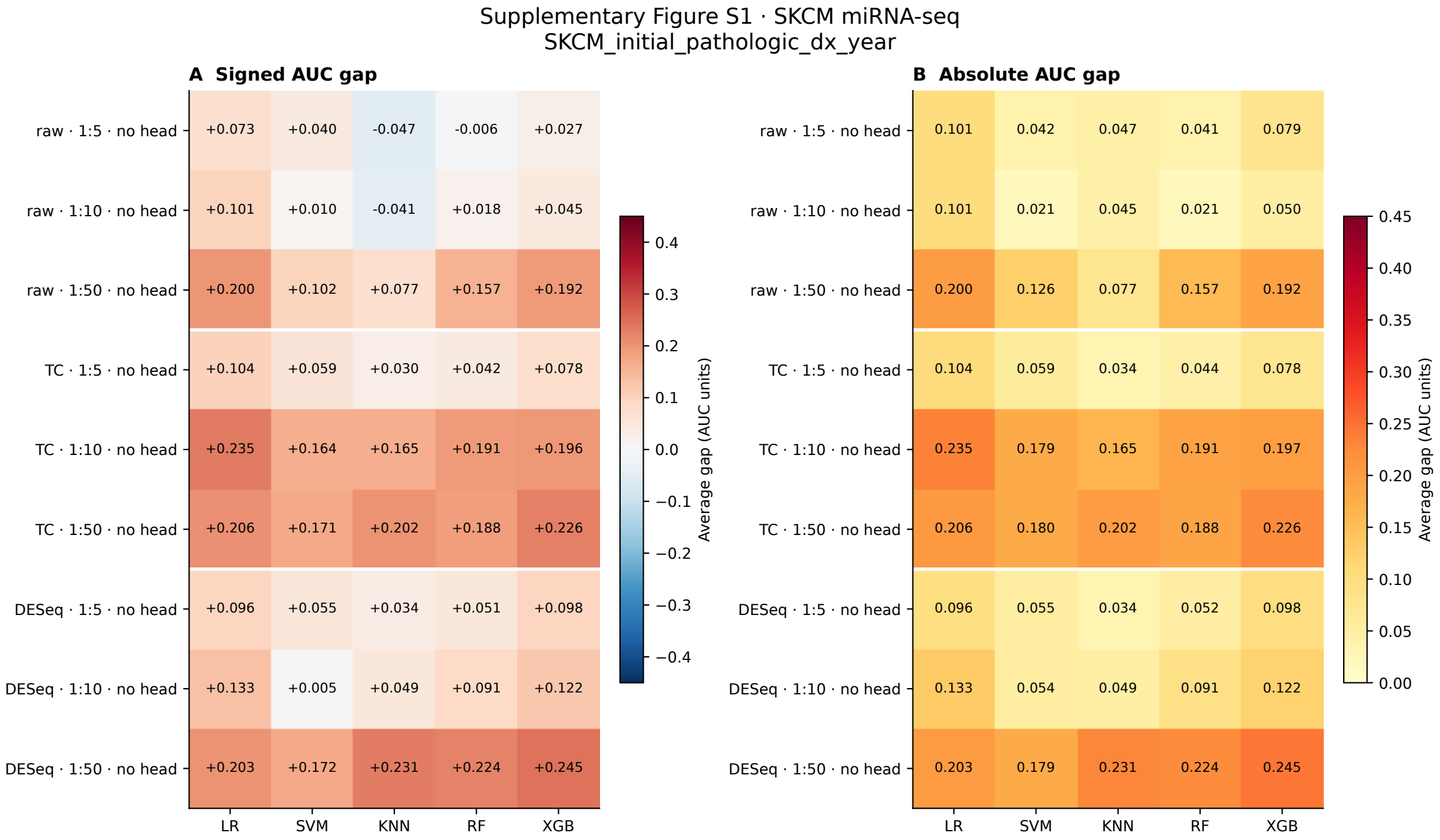

Supplementary Figure S1 · PRAD miRNA-seq
PRAD_icd_o_3_histology
A Signed AUC gap
B Absolute AUC gap
raw · 1:5 · no head
raw · 1:10 · no head
raw · 1:50 · no head
TC · 1:5 · no head
TC · 1:10 · no head
TC · 1:50 · no head
DESeq · 1:5 · no head
DESeq · 1:10 · no head
DESeq · 1:50 · no head
-0.092 -0.144 -0.193 -0.140 -0.110
-0.110 -0.154 -0.226 -0.166 -0.130
-0.298 -0.351 -0.388 -0.364 -0.341
-0.116 -0.145 -0.215 -0.134 -0.121
-0.031 -0.065 -0.118 -0.046 -0.029
-0.196 -0.223 -0.271 -0.211 -0.221
-0.215 -0.245 -0.327 -0.272 -0.242
-0.253 -0.294 -0.382 -0.326 -0.287
-0.191 -0.206 -0.293 -0.207 -0.200
0.092 0.146 0.193 0.140 0.114
0.110 0.158 0.226 0.166 0.135
0.298 0.365 0.388 0.364 0.341
0.116 0.149 0.216 0.134 0.124
0.031 0.066 0.118 0.046 0.032
0.196 0.223 0.271 0.211 0.221
0.215 0.250 0.327 0.272 0.242
0.253 0.296 0.382 0.326 0.287
0.191 0.207 0.293 0.207 0.200
LR SVM KNN RF XGB
Average gap (AUC units)
0.4 0.3 0.2 0.1 0.0 −0.1 −0.2 −0.3 −0.4
0.45 0.40 0.35 0.30 0.25 0.20 0.15 0.10 0.05 0.00

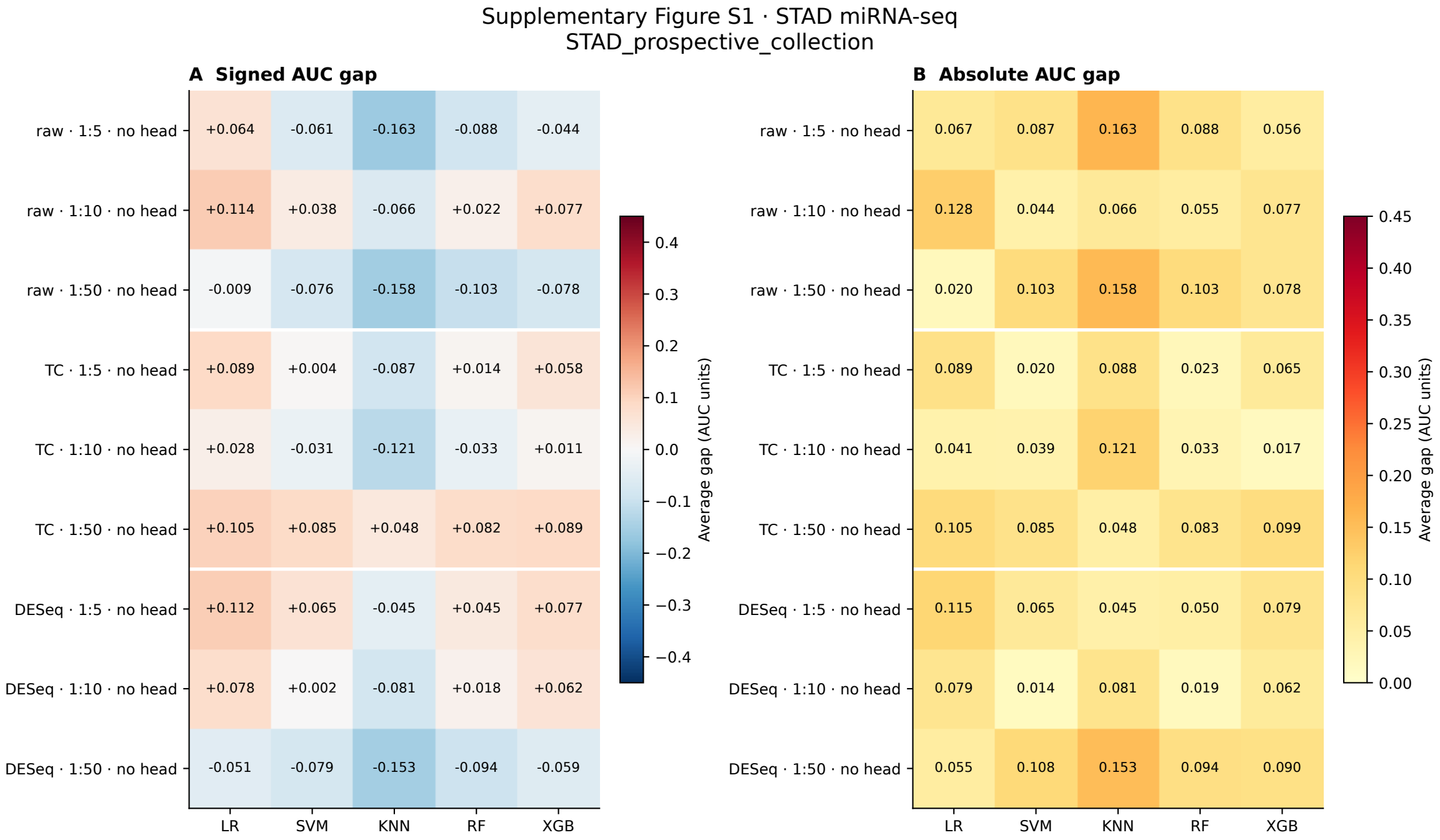
Supplementary Figure S1 · READ miRNA-seq
READ_tumor_status

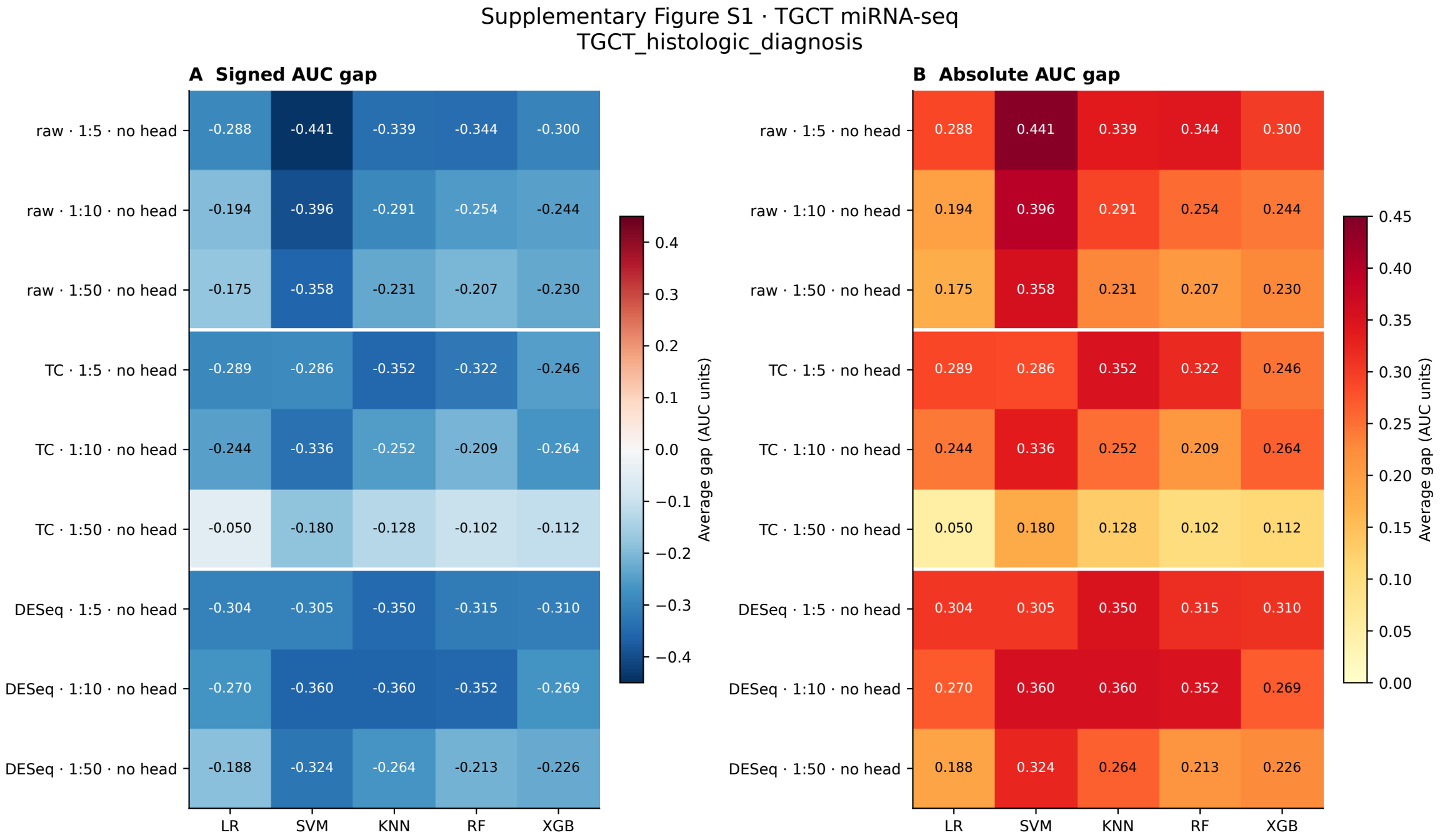


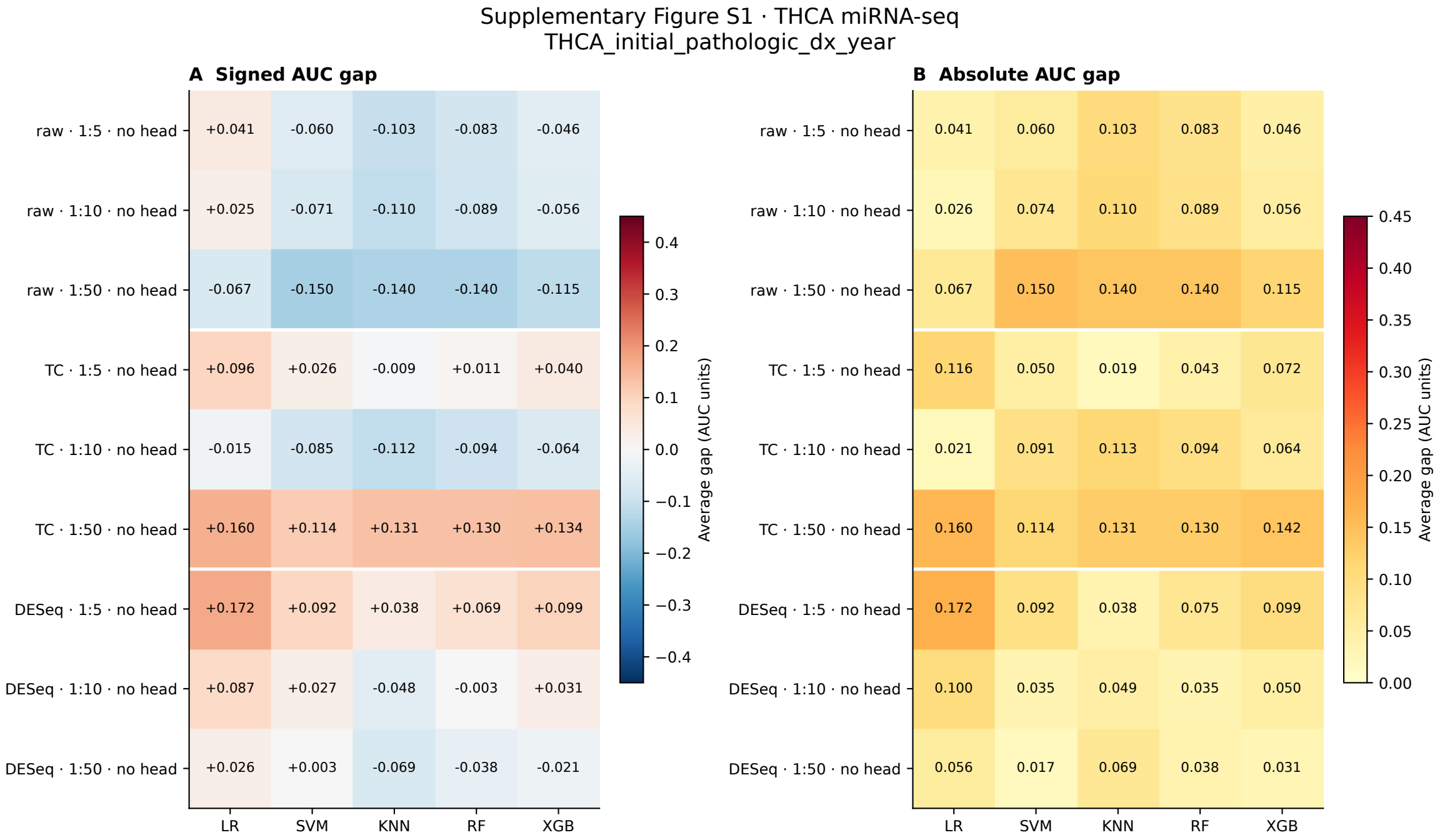
Supplementary Figure S1 · SARC miRNA-seq
SARC_primary_pathology_histological_type



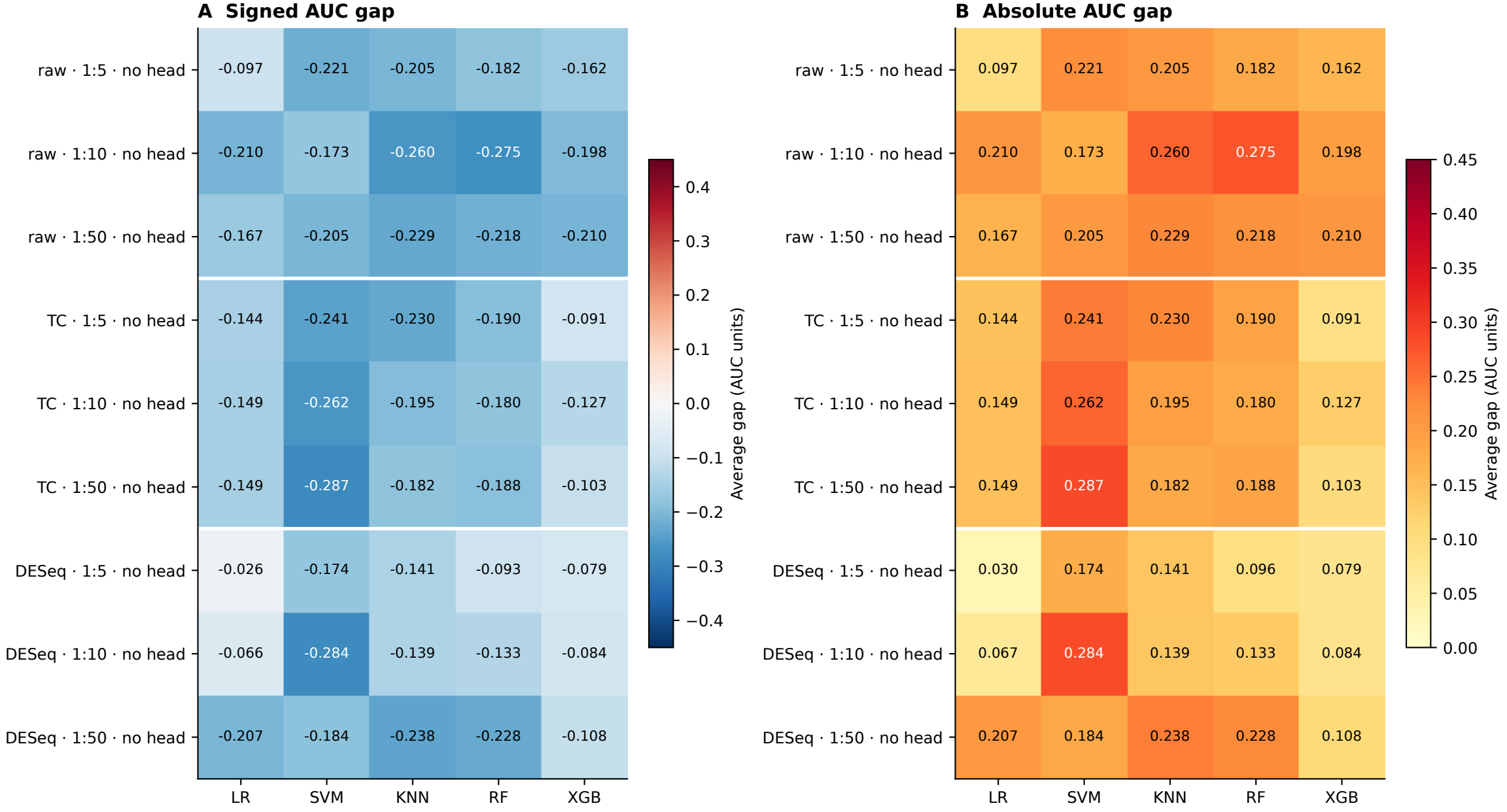

Supplementary Figure S1 · SKCM miRNA-seq
SKCM_initial_pathologic_dx_year
A Signed AUC gap
B Absolute AUC gap
raw · 1:5 · no head
raw · 1:10 · no head
raw · 1:50 · no head
TC · 1:5 · no head
TC · 1:10 · no head
TC · 1:50 · no head
DESeq · 1:5 · no head
DESeq · 1:10 · no head
DESeq · 1:50 · no head
+0.073 +0.040 -0.047 -0.006 +0.027
+0.101 +0.010 -0.041 +0.018 +0.045
+0.200 +0.102 +0.077 +0.157 +0.192
+0.104 +0.059 +0.030 +0.042 +0.078
+0.235 +0.164 +0.165 +0.191 +0.196
+0.206 +0.171 +0.202 +0.188 +0.226
+0.096 +0.055 +0.034 +0.051 +0.098
+0.133 +0.005 +0.049 +0.091 +0.122
+0.203 +0.172 +0.231 +0.224 +0.245
0.101 0.042 0.047 0.041 0.079
0.101 0.021 0.045 0.021 0.050
0.200 0.126 0.077 0.157 0.192
0.104 0.059 0.034 0.044 0.078
0.235 0.179 0.165 0.191 0.197
0.206 0.180 0.202 0.188 0.226
0.096 0.055 0.034 0.052 0.098
0.133 0.054 0.049 0.091 0.122
0.203 0.179 0.231 0.224 0.245
LR SVM KNN RF XGB
Average gap (AUC units)
0.4 0.3 0.2 0.1 0.0 −0.1 −0.2 −0.3 −0.4
0.45 0.40 0.35 0.30 0.25 0.20 0.15 0.10 0.05 0.00


Supplementary Figure S1 · STAD miRNA-seq
STAD_prospective_collection
A Signed AUC gap
B Absolute AUC gap
raw · 1:5 · no head
raw · 1:10 · no head
raw · 1:50 · no head
TC · 1:5 · no head
TC · 1:10 · no head
TC · 1:50 · no head
DESeq · 1:5 · no head
DESeq · 1:10 · no head
DESeq · 1:50 · no head
+0.064 -0.061 -0.163 -0.088 -0.044
+0.114 +0.038 -0.066 +0.022 +0.077
-0.009 -0.076 -0.158 -0.103 -0.078
+0.089 +0.004 -0.087 +0.014 +0.058
+0.028 -0.031 -0.121 -0.033 +0.011
+0.105 +0.085 +0.048 +0.082 +0.089
+0.112 +0.065 -0.045 +0.045 +0.077
+0.078 +0.002 -0.081 +0.018 +0.062
-0.051 -0.079 -0.153 -0.094 -0.059
0.067 0.087 0.163 0.088 0.056
0.128 0.044 0.066 0.055 0.077
0.020 0.103 0.158 0.103 0.078
0.089 0.020 0.088 0.023 0.065
0.041 0.039 0.121 0.033 0.017
0.105 0.085 0.048 0.083 0.099
0.115 0.065 0.045 0.050 0.079
0.079 0.014 0.081 0.019 0.062
0.055 0.108 0.153 0.094 0.090
LR SVM KNN RF XGB
Average gap (AUC units)
0.4 0.3 0.2 0.1 0.0 −0.1 −0.2 −0.3 −0.4
0.45 0.40 0.35 0.30 0.25 0.20 0.15 0.10 0.05 0.00

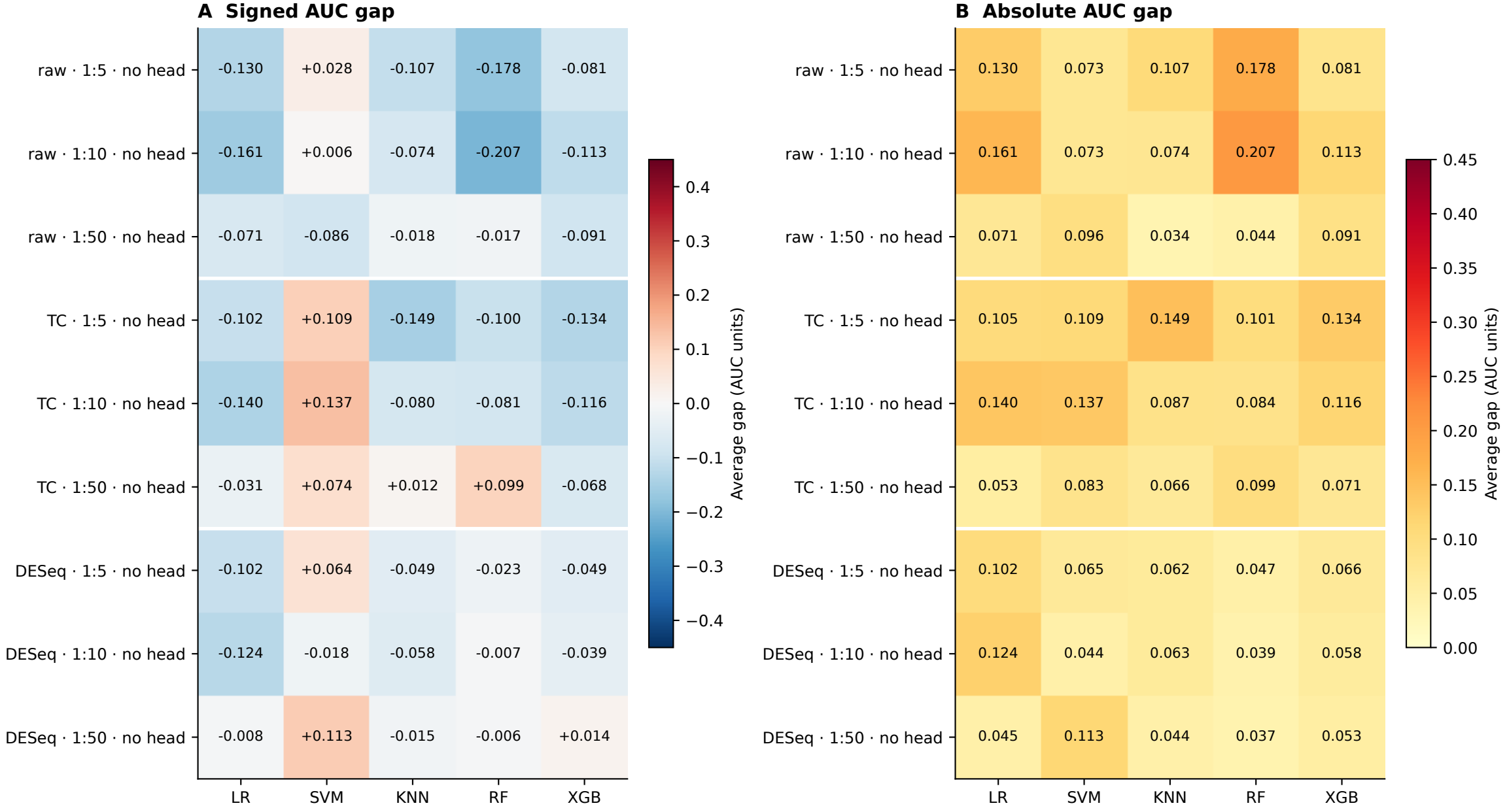

Supplementary Figure S1 · TGCT miRNA-seq
TGCT_histologic_diagnosis
A Signed AUC gap
B Absolute AUC gap
raw · 1:5 · no head
raw · 1:10 · no head
raw · 1:50 · no head
TC · 1:5 · no head
TC · 1:10 · no head
TC · 1:50 · no head
DESeq · 1:5 · no head
DESeq · 1:10 · no head
DESeq · 1:50 · no head
-0.288 -0.441 -0.339 -0.344 -0.300
-0.194 -0.396 -0.291 -0.254 -0.244
-0.175 -0.358 -0.231 -0.207 -0.230
-0.289 -0.286 -0.352 -0.322 -0.246
-0.244 -0.336 -0.252 -0.209 -0.264
-0.050 -0.180 -0.128 -0.102 -0.112
-0.304 -0.305 -0.350 -0.315 -0.310
-0.270 -0.360 -0.360 -0.352 -0.269
-0.188 -0.324 -0.264 -0.213 -0.226
0.288 0.441 0.339 0.344 0.300
0.194 0.396 0.291 0.254 0.244
0.175 0.358 0.231 0.207 0.230
0.289 0.286 0.352 0.322 0.246
0.244 0.336 0.252 0.209 0.264
0.050 0.180 0.128 0.102 0.112
0.304 0.305 0.350 0.315 0.310
0.270 0.360 0.360 0.352 0.269
0.188 0.324 0.264 0.213 0.226
LR SVM KNN RF XGB
Average gap (AUC units)
0.4 0.3 0.2 0.1 0.0 −0.1 −0.2 −0.3 −0.4
0.45 0.40 0.35 0.30 0.25 0.20 0.15 0.10 0.05 0.00

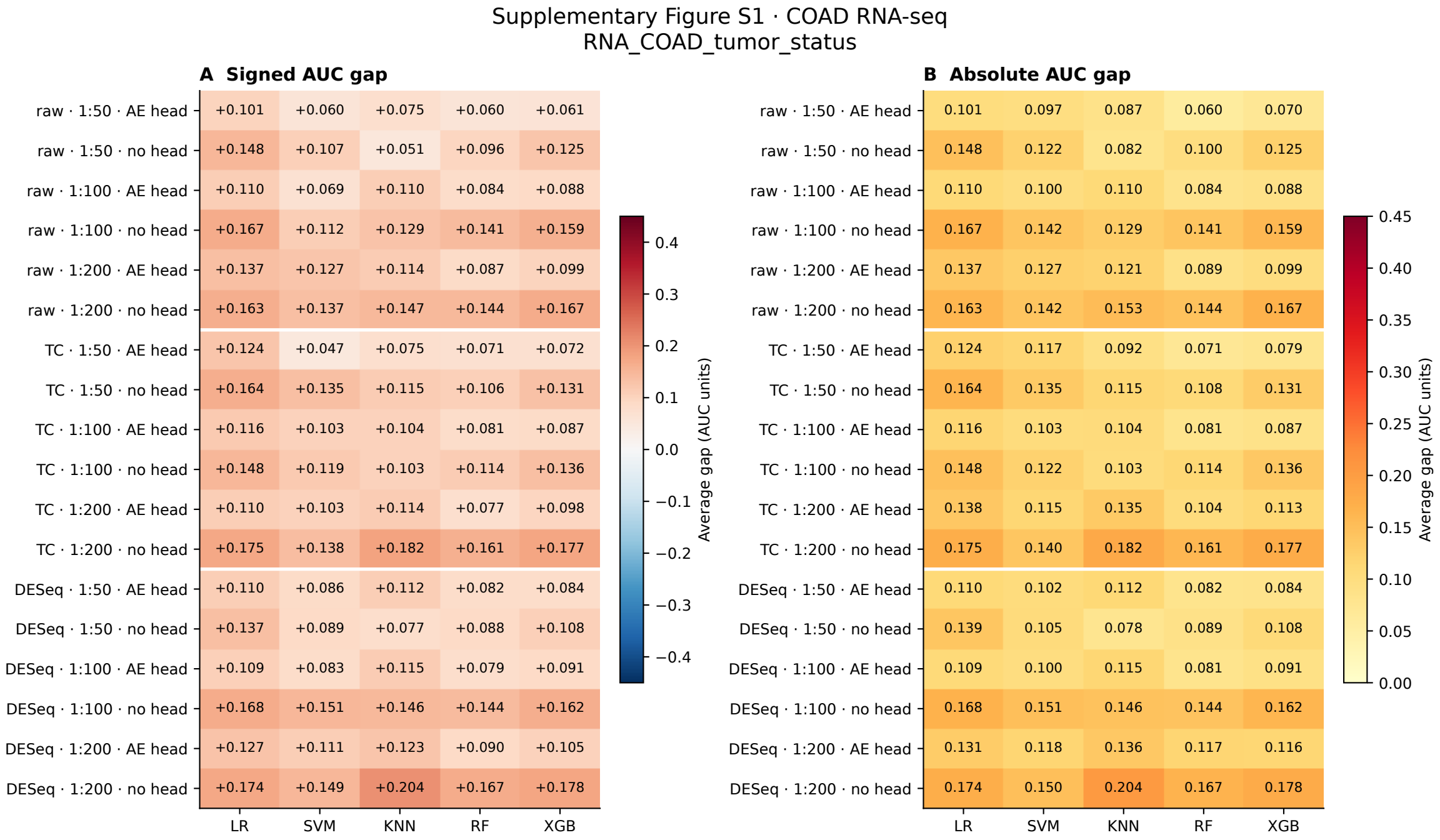

Supplementary Figure S1 · THCA miRNA-seq
THCA_initial_pathologic_dx_year
A Signed AUC gap
B Absolute AUC gap
raw · 1:5 · no head
raw · 1:10 · no head
raw · 1:50 · no head
TC · 1:5 · no head
TC · 1:10 · no head
TC · 1:50 · no head
DESeq · 1:5 · no head
DESeq · 1:10 · no head
DESeq · 1:50 · no head
+0.041 -0.060 -0.103 -0.083 -0.046
+0.025 -0.071 -0.110 -0.089 -0.056
-0.067 -0.150 -0.140 -0.140 -0.115
+0.096 +0.026 -0.009 +0.011 +0.040
-0.015 -0.085 -0.112 -0.094 -0.064
+0.160 +0.114 +0.131 +0.130 +0.134
+0.172 +0.092 +0.038 +0.069 +0.099
+0.087 +0.027 -0.048 -0.003 +0.031
+0.026 +0.003 -0.069 -0.038 -0.021
0.041 0.060 0.103 0.083 0.046
0.026 0.074 0.110 0.089 0.056
0.067 0.150 0.140 0.140 0.115
0.116 0.050 0.019 0.043 0.072
0.021 0.091 0.113 0.094 0.064
0.160 0.114 0.131 0.130 0.142
0.172 0.092 0.038 0.075 0.099
0.100 0.035 0.049 0.035 0.050
0.056 0.017 0.069 0.038 0.031
LR SVM KNN RF XGB
Average gap (AUC units)
0.4 0.3 0.2 0.1 0.0 −0.1 −0.2 −0.3 −0.4
0.45 0.40 0.35 0.30 0.25 0.20 0.15 0.10 0.05 0.00

Supplementary Figure S1 · THYM miRNA-seq
THYM_primary_pathology_history_myasthenia_gravis



Supplementary Figure S1 · UCS miRNA-seq
UCS_primary_pathology_total_pelv_lnr




Supplementary Figure S1 · COAD RNA-seq
RNA_COAD_tumor_status
A Signed AUC gap
B Absolute AUC gap
raw · 1:50 · AE head +0.101 +0.060 +0.075 +0.060 +0.061
raw · 1:50 · no head +0.148 +0.107 +0.051 +0.096 +0.125
raw · 1:100 · AE head +0.110 +0.069 +0.110 +0.084 +0.088
raw · 1:100 · no head +0.167 +0.112 +0.129 +0.141 +0.159
raw · 1:200 · AE head +0.137 +0.127 +0.114 +0.087 +0.099
raw · 1:200 · no head +0.163 +0.137 +0.147 +0.144 +0.167
TC · 1:50 · AE head +0.124 +0.047 +0.075 +0.071 +0.072
TC · 1:50 · no head +0.164 +0.135 +0.115 +0.106 +0.131
TC · 1:100 · AE head +0.116 +0.103 +0.104 +0.081 +0.087
TC · 1:100 · no head +0.148 +0.119 +0.103 +0.114 +0.136
TC · 1:200 · AE head +0.110 +0.103 +0.114 +0.077 +0.098
TC · 1:200 · no head +0.175 +0.138 +0.182 +0.161 +0.177
DESeq · 1:50 · AE head +0.110 +0.086 +0.112 +0.082 +0.084
DESeq · 1:50 · no head +0.137 +0.089 +0.077 +0.088 +0.108
DESeq · 1:100 · AE head +0.109 +0.083 +0.115 +0.079 +0.091
DESeq · 1:100 · no head +0.168 +0.151 +0.146 +0.144 +0.162
DESeq · 1:200 · AE head +0.127 +0.111 +0.123 +0.090 +0.105
DESeq · 1:200 · no head +0.174 +0.149 +0.204 +0.167 +0.178
LR SVM KNN RF XGB
0.4 0.3 0.2 0.1 0.0 −0.1 −0.2 −0.3 −0.4
Average gap (AUC units)
raw · 1:50 · AE head 0.101 0.097 0.087 0.060 0.070
raw · 1:50 · no head 0.148 0.122 0.082 0.100 0.125
raw · 1:100 · AE head 0.110 0.100 0.110 0.084 0.088
raw · 1:100 · no head 0.167 0.142 0.129 0.141 0.159
raw · 1:200 · AE head 0.137 0.127 0.121 0.089 0.099
raw · 1:200 · no head 0.163 0.142 0.153 0.144 0.167
TC · 1:50 · AE head 0.124 0.117 0.092 0.071 0.079
TC · 1:50 · no head 0.164 0.135 0.115 0.108 0.131
TC · 1:100 · AE head 0.116 0.103 0.104 0.081 0.087
TC · 1:100 · no head 0.148 0.122 0.103 0.114 0.136
TC · 1:200 · AE head 0.138 0.115 0.135 0.104 0.113
TC · 1:200 · no head 0.175 0.140 0.182 0.161 0.177
DESeq · 1:50 · AE head 0.110 0.102 0.112 0.082 0.084
DESeq · 1:50 · no head 0.139 0.105 0.078 0.089 0.108
DESeq · 1:100 · AE head 0.109 0.100 0.115 0.081 0.091
DESeq · 1:100 · no head 0.168 0.151 0.146 0.144 0.162
DESeq · 1:200 · AE head 0.131 0.118 0.136 0.117 0.116
DESeq · 1:200 · no head 0.174 0.150 0.204 0.167 0.178
LR SVM KNN RF XGB
0.45 0.40 0.35 0.30 0.25 0.20 0.15 0.10 0.05 0.00
Average gap (AUC units)

Supplementary Figure S1 · SKCM RNA-seq
RNA_SKCM_breslow_thickness_at_diagnosis

A Signed AUC gap

	LR	SVM	KNN	RF	XGB
raw · 1:50 · AE head	+0.140	+0.149	+0.118	+0.099	+0.084
raw · 1:50 · no head	+0.227	+0.159	+0.099	+0.151	+0.182
raw · 1:100 · AE head	+0.139	+0.128	+0.137	+0.110	+0.089
raw · 1:100 · no head	+0.212	+0.109	+0.066	+0.132	+0.139
raw · 1:200 · AE head	+0.143	+0.136	+0.167	+0.132	+0.133
raw · 1:200 · no head	+0.193	+0.123	+0.044	+0.123	+0.188
TC · 1:50 · AE head	+0.133	+0.143	+0.124	+0.103	+0.100
TC · 1:50 · no head	+0.217	+0.179	+0.096	+0.147	+0.192
TC · 1:100 · AE head	+0.097	+0.166	+0.128	+0.077	+0.090
TC · 1:100 · no head	+0.227	+0.183	+0.084	+0.162	+0.230
TC · 1:200 · AE head	+0.129	+0.163	+0.147	+0.076	+0.132
TC · 1:200 · no head	+0.249	+0.236	+0.151	+0.204	+0.263
DESeq · 1:50 · AE head	+0.104	+0.114	+0.124	+0.079	+0.072
DESeq · 1:50 · no head	+0.223	+0.206	+0.101	+0.156	+0.215
DESeq · 1:100 · AE head	+0.136	+0.136	+0.163	+0.117	+0.130
DESeq · 1:100 · no head	+0.245	+0.245	+0.195	+0.209	+0.264
DESeq · 1:200 · AE head	+0.144	+0.154	+0.176	+0.133	+0.122
DESeq · 1:200 · no head	+0.152	+0.143	+0.065	+0.103	+0.164

Average gap (AUC units): 0.4, 0.3, 0.2, 0.1, 0.0, −0.1, −0.2, −0.3, −0.4

B Absolute AUC gap

	LR	SVM	KNN	RF	XGB
raw · 1:50 · AE head	0.148	0.149	0.126	0.110	0.097
raw · 1:50 · no head	0.229	0.159	0.099	0.158	0.182
raw · 1:100 · AE head	0.151	0.128	0.147	0.113	0.105
raw · 1:100 · no head	0.214	0.109	0.066	0.132	0.150
raw · 1:200 · AE head	0.145	0.136	0.167	0.135	0.133
raw · 1:200 · no head	0.201	0.123	0.057	0.138	0.188
TC · 1:50 · AE head	0.144	0.143	0.135	0.112	0.114
TC · 1:50 · no head	0.228	0.179	0.116	0.161	0.192
TC · 1:100 · AE head	0.144	0.166	0.154	0.114	0.131
TC · 1:100 · no head	0.235	0.183	0.099	0.172	0.230
TC · 1:200 · AE head	0.158	0.163	0.174	0.135	0.132
TC · 1:200 · no head	0.251	0.236	0.163	0.212	0.263
DESeq · 1:50 · AE head	0.123	0.115	0.137	0.083	0.079
DESeq · 1:50 · no head	0.234	0.206	0.113	0.164	0.215
DESeq · 1:100 · AE head	0.145	0.136	0.178	0.123	0.130
DESeq · 1:100 · no head	0.247	0.245	0.195	0.209	0.264
DESeq · 1:200 · AE head	0.150	0.154	0.180	0.133	0.122
DESeq · 1:200 · no head	0.182	0.143	0.101	0.116	0.164

Average gap (AUC units): 0.45, 0.40, 0.35, 0.30, 0.25, 0.20, 0.15, 0.10, 0.05, 0.00